

Time to Eat - A Personalized Circadian Eating Schedule Leads to Weight Loss Without Calorie Restriction: A Randomized Controlled Trial

Isabell Wilming, Jana Tuschewski, Jessie M Osterhaus, Theresa JG Bringmann, Anisja Hühne-Landgraf, Dominic Landgraf 

Circadian Biology Group, Department of Molecular Neurobiology, Clinic of Psychiatry and Psychotherapy, University Hospital, Ludwig Maximilian University, Munich, Germany • Munich Medical Research School, Ludwig Maximilian University, Munich, Germany

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Abstract

Objective

Many weight loss strategies are based on the restriction of calories or certain foods. Here, we tested a weight loss intervention based solely on increasing the regularity of meals to allow the circadian system to optimally prepare food metabolism for these times.

Participants & Methods

In a two-group, single center randomized-controlled single-blind study (pre-registration DRKS00021419) with participants aged 18-65 years and BMI ≥ 22 kg/m², we used a smartphone application to identify the times at which each participant eats particularly frequently and asked participants of the experimental group to restrict their meals to only these times for six weeks. Control participants received sham treatment. Primary outcome was self-reported body weight/BMI and secondary outcome the well-being of participants.

Results

Of 148 participants entering the study, 121 were randomized and of these 100 (control: 33, experimental: 67) finished the study. Our results show that the more regular the meals of participants of the experimental group became, the more weight/BMI they lost, averaging 2.62 kg (0.87 kg/m²); $p < 0.0001$ (BMI: $p < 0.0001$) compared to an insignificant weight loss of 0.56 kg (0.20 kg/m²) in the control group; $p = 0.0918$ (BMI: $p = 0.0658$). Strikingly, weight loss was not related to changes in self-reported calories, food composition, and other food-related factors. Additionally, physical and mental well-being improved significantly.

Conclusion

In summary, increasing the regularity of meals causes participants to lose excess body weight and improves overall well-being.

Highlights

- Individual optimal times for meals are determined via an app-based meal diary.
- Generation of a structure plan for mealtimes are adjusted to individual circadian clocks.
- Following this plan, participants lost an average body weight of 2.6 kg over six weeks.
- Weight loss is achieved without changes in self-reported food quantity or composition.
- Regular mealtimes contribute to the improvement of the general well-being.

eLife assessment

This study investigates a dietary intervention that employs a smartphone app to promote meal regularity, which may be **useful**. Despite no observed changes in caloric intake, the authors report significant weight loss. While the concept is very interesting and deserves to be studied due to its potential clinical relevance, the study's rigor needs to be improved, and is currently considered **inadequate**, notably for its reliance on self-reported food intake, a highly unreliable way to assess food intake. Additionally, the study theorizes that the intervention resets the circadian clock, but the study needs more reliable methods for assessing circadian rhythms, such as actigraphy.

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1. Introduction

Circadian clocks induce endogenous 24-hour rhythms in the expression of more than half of all genes across all tissues [1, 2], allowing the body to anticipate daily changes between day and night and to prepare and harmonize physiological processes accordingly. However, this requires exposure to very regular environmental time cues, so-called *Zeitgebers*, such as food. Thus, on the one hand, the circadian system contributes to the optimization of food metabolism; on the other hand, the intake of food serves as a potent *Zeitgeber*. Both mouse and human studies have shown the negative consequences of circadian disturbances on metabolism, body weight [3–11], and also mental health [12–14] and demonstrated that restriction of meals to certain time spans of the day can counteract these harmful effects [15–19]. However, previous human studies merely established rough periods during which participants could still eat irregularly.

We expect that the success of previous interventions can be further improved by determining each participant's eating time profile and creating a personalized meal plan accordingly, even if participants continue to eat large meals.

Therefore, the primary aim is to individually determine and set optimal times for food intake and to correlate the reduction of mealtime variability with alterations in body weight. For control subjects, we specified a long 18-h window during which they could continue to eat irregularly. We include participants of Body Mass Index (BMI) classifications normal to extremely obese to test whether participants who exceed normal BMIs particularly benefit from the intervention. Because of the impact of circadian clocks on other body functions, we further hypothesize that regular food intake contributes to an overall improvement in well-being, such as the subjective feeling of general health, sleep quality, affective state, and self-efficacy. Since the program focuses on meal times and, unlike many other dietary programs, explicitly does not aim to limit calories or certain types of food, we have named it “Time to Eat” ([Fig. S1A](#)).

2. Results

2.1. Participants

A total of 148 participants were recruited. Of these, 121 were randomly allocated to the experimental group (EG) ($n=79$) and the control group (CG) ($n=42$) ([Fig. S1B](#)), which had non-significantly different mean BMIs of 25.9 (SD: ± 4.138) (CG) and 27.5 (SD: ± 4.627) (EG) ([Tab. S1](#)). At the end of the intervention (T2), data were available from 100 participants (EG: $n=67$, CG: $n=33$). Their baseline data indicate that participants in both groups were, on average, overweight according to BMI classification, with the included CG participants having a BMI 25.5 (SD: ± 3.827) and being slightly less overweight, on average, than the analyzed EG participants with a BMI of 27.4 (SD: ± 4.616) ([Tab. S1](#)). There were no significant BMI differences between completers and drop-outs in either group ([Fig. S2A](#), [Tab. S2](#)). Reasons for drop-outs were, according to self-report, illness/accident during the study and unwillingness or failure to adhere to fixed meal times.

2.2. Improvement of meal time regularity

All study participants were asked to record each caloric event during a 14-day exploration and a six-week intervention phase ([Fig. 1A](#), [B](#), [S1C](#)). During the exploration and intervention phases, a total of 13,838 and 34,564 caloric events were recorded from completers, respectively. Most EG participants were assessed as having three eating times ($n=53$), some reported eating four meals ($n=13$), and one had two main meals per day. The average MTVS for breakfast, lunch, dinner, and total meals of the exploration phase of both groups was around 4, which corresponds to a daily deviation of ~ 120 min for each meal. In the EG group, mean scores improved to less than 2 during the intervention phase, corresponding to a mean deviation of less than ± 30 min ([Fig. 1C](#), [D](#), [Tab. S3](#)). In contrast, in the CG group, there was no improvement but some significant increase of the MTVS.

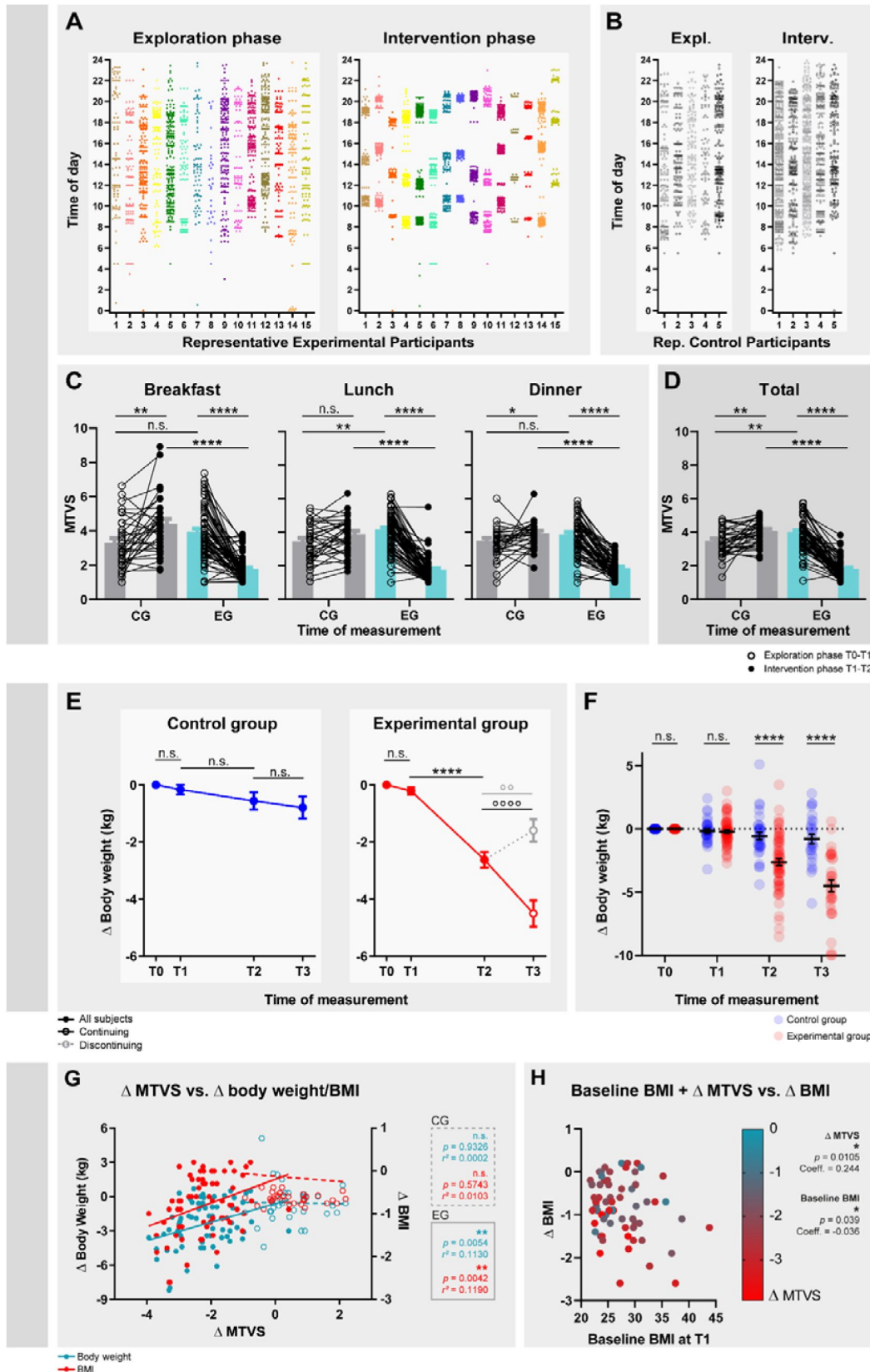


Fig. 1

Improvement in regularity of meals, body weight changes, and correlation between loss of body weight and regularity of eating meals. (A, B) Times of day of caloric events of 15 representative participants of the EG (A) and the CG (B) during the two-week exploration phase (left) and the six-week intervention phase (right). Each point represents a caloric event. (C, D) Change in MTVS of participants in the CG (gray) and EG (cyan) of breakfast, lunch, and dinner (C) and of all meals combined (D), including snacks, if snacks were taken. 2-way repeated-measures ANOVA with Bonferroni post hoc test comparing CG and EG during the exploration or intervention phase and T0-T1 and T1-T2 within groups, * $p \leq 0.05$, ** $p \leq 0.01$, **** $p \leq 0.0001$. CG: $n=33$; EG: $n=64-67$, missing n -values in the EG are due to the different number of meals for each participant. Detailed statistics in [Tables S3](#). (E) Within group differences of body weight during the study. Body weight was collected at different time points, at the beginning of the study (T0), after the exploration phase (T1), after the intervention phase (T2), and after the follow-up phase (T3). During the follow-up phase, 34 EG participants voluntarily continued the intervention (continuing; black line and open circle) while 33 participants no longer followed the intervention (discontinuing; gray, dashed line and open circle). Only three control subjects continued their intervention in the follow-up phase, which is why a division into continuing and discontinuing was omitted in this group. Data are normalized so that all baseline data at T0 equals 0 to represent change. Differences of all CG and EG participants between T0, T1, and T2 were calculated with a 1-way ANOVA, as were differences between T2 and T3 of CG participants (Bonferroni post-hoc test, **** $p \leq 0.0001$). Differences between T2 and T3 of continuing and discontinuing EG participants were calculated with a paired t-test ** $p \leq 0.01$, *** $p \leq 0.0001$. Detailed statistics in [Table S4](#). (F) Between group differences of BMI from T0-T3. Mixed-effect model with Bonferroni post-hoc test comparing CG and EG at different phases during the study, **** $p \leq 0.0001$. Detailed statistics in [Table S5](#). (G) Significant correlation between body weight/BMI and MTVS. The more participants improved their regularity of meals (decreasing MTVS), the more weight/BMI they lost. Linear regression, CG: $n=33$; EG: $n=67$, ** $p \leq 0.01$. Detailed statistics in [Tab. S6](#). (H) Significant correlation between change in BMI, baseline BMI, and improvement in MTVS. Individuals with higher BMI benefit more from similar improvements in MTVS (shades of data points) than individuals with lower BMI in EG participants. Multiple regression, $n=67$, * $p \leq 0.05$, ** $p \leq 0.01$. Detailed statistics in [Tab. S9](#).

2.3. Weight change

During the exploration phase, when no instructions on eating times were given, CG and EG participants had a statistically non-significant average loss of 0.17 kg (0.06 kg/m²) and 0.21 kg (0.06 kg/m²), respectively ([Figs. 1E](#), [S2B-F](#), [Tabs. S4, 5](#)). However, during the intervention phase, when EG participants took their meals more regularly, they lost an average of about 2.6 kg (95% CI [-2.906, -1.915]) and 0.81 kg/m² (95% CI [-0.9729, -0.6420]) ([Figs. 1E](#), [S2B-E](#), [Tabs. S4](#), [5](#)), which translates to an average weight loss of 0.40 kg (0.145 kg/m²) per week during the intervention. In contrast, weight and BMI changes of -0.39 kg (95% CI [-0.9125, 0.1307]) and -0.14 kg/m² (95% CI [-0.3152, 0.0304]), respectively, were not significant in the CG ([Figs. 1E](#), [S2B, C, F](#), [Tabs. S4, 5](#)), resulting in a significant difference in weight loss between groups at the end of the intervention ([Figs. 1F](#), [S2C](#), [Tab. S5](#)). The inclusion of the last available weight and BMI data of the drop-outs in an end-point analysis did not lead to any notable change in this result ([Fig. S2G](#), [Tab. S5](#)).

Four weeks after the intervention ended, all participants were contacted again for a follow-up assessment. 34 of the 67 EG participants reported to have maintained the regularity of their meals and continued to lose body weight significantly, on average another 1.12 kg (95% CI [-1.479, -0.7626]) and 0.36 kg/m² (95% CI [-0.4907, -0.2457]), respectively ([Figs. 1E](#), [S2B, D, E](#), [Tabs. S4, 5](#)). In contrast, participants who reported to have resumed irregular eating gained a significant average of 0.48 kg (95% CI [0.1984, 0.7616]) and 0.17 kg/m² (95% CI [0.07434, 0.2603]), respectively ([Fig. 1E](#), [S2B](#), [D](#), [E](#), [Tab. S4](#)).

2.4. Relation between weight change and improvement in the regularity of meals

To examine the extent to which EG participants' weight loss was effectively related to improvements in meal regularity, we correlated changes in body weight and BMI with individual MTVS changes. Consistent with our hypothesis, a significant relationship exists between weight loss and improvement in regularity in the EG (**Fig. 1G**, **Tab. S6**). A breakdown of meals shows that increasing the regularity of lunch and dinner has the greatest effects, with no significant relationship between increasing the regularity of breakfast and weight loss (**Fig. S2H**, **Tab. S6**). In contrast, in the CG, there were no or little improvements in regularity and, accordingly, no correlation with weight/BMI changes.

Further analyses revealed that all MTVS up to 3, which corresponds to a time window for meals of up to ± 45 min, resulted in comparably effective weight loss, whereas the efficiency of weight loss decreases significantly with an MTVS greater than 3 (**Tab. S7**).

2.5. Relation between weight change and baseline BMI

A weight loss intervention which is particularly based on metabolic optimization is likely to be most effective for individuals with exceeded body weight. In fact, the higher the baseline BMI of EG participants at T1, the greater the weight loss (**Fig. S2I**, **Tab. S6**). Stratifying the subjects by baseline BMI, including the control group, confirms this result, but also shows that in all strata the BMI of EG subjects decreased more than that of CG subjects, even if they had the same initial BMI (**Fig. S2J**, **Tab. S8**). Multiple regression shows that the reduction in MTVS and baseline BMI together predict the weight loss of the participants (**Fig. 1H**, **Tab. S9**). These data show that regularity in meal intake can lead to a significant loss of body weight and suggest that this effect is especially pronounced in individuals with a higher BMI.

2.6. Relation between weight change and self-reported food quantity and composition

By restricting meals to specific times and shortening the time window in which food is eaten, it is feasible that EG participants ate less or changed food composition during the intervention phase and therefore lost weight. However, on average, the reported food energy ingested by CG and EG participants did not substantially change during the study (**Fig. 2A**, **S2K**, **L**, **Tab. S3**, **6**), apart from a minor average reduction of ~ 350 kJ (95% CI [-627.6, -67.67]) per day in the EG (**Fig. S2K**, **Tab. S3**, **6**), and did not differ between CG and the EG. As they are rather small, these differences do not add up to a significant reduction in the total average energy consumed over the course of the study. Accordingly, the determined cumulative average energy intake during the intervention phase did not differ on any day from the estimated energy intake based on an extrapolation of the cumulative food consumption during the exploration phase, neither in the CG nor in the EG group (**Fig. 2B**). Nevertheless, there were some participants who reported having consumed fewer calories during the intervention phase, while some also reported having ingested more calories. However, consistent with our hypothesis that high regularity of mealtimes optimizes metabolism of similar amounts of food, there is no relationship between reported changes in average daily and cumulative energy intake and change in body weight/BMI of either CG or EG participants (**Fig. 2C**, **D**, **Tab. S6**). Likewise, the composition of the ingested foods did not change considerably in either group, and participants reported a normal distribution of the macronutrients of fat, carbohydrate, and protein in both phases of the study (**Fig. 2E**, **S2M**). Also, there was no correlation between weight change and change in reported average daily and cumulative macronutrient intake (**Fig. 2D**, **S2N**, **Tab. S6**).

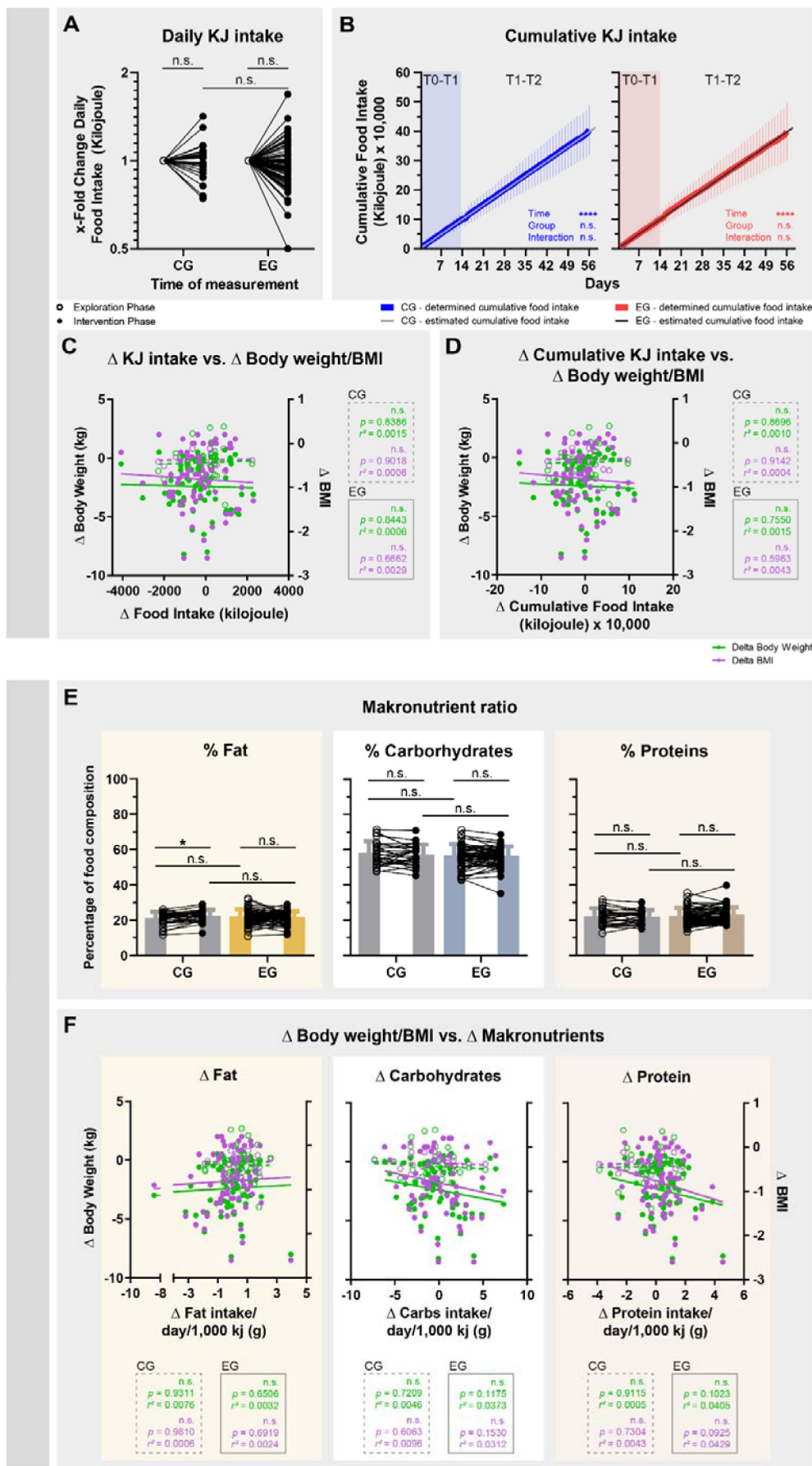


Fig. 2

Changes in body weight/BMI are not related to changes in reported caloric intake and composition of macronutrients. Changes in reported daily caloric intake (A), cumulative caloric intake (B) and food composition (E) and their relationship to weight/BMI changes (C, D, and F). Data in (A) are normalized so that all baseline data of the exploration phase equal 1. To represent the x-fold change, data from the intervention phase are divided by data from the exploration phase of the corresponding participant. (A, E) 2-way repeated-measures ANOVA with Bonferroni post hoc test comparing CG and EG during the exploration or intervention phase and T0-T1 and T1-T2 within groups, * $p \leq 0.05$. Detailed statistics in [Tables S3](#). (B) Mixed-effects model with Bonferroni post hoc test comparing determined and estimated cumulative energy intake. The estimated energy intake is formed as an assumed continuation of the average daily intake during the exploration phase. The Bonferroni post hoc test showed no significant differences on any study day. (C, D, and F) Linear regression, CG: $n=30$; EG: $n=67$ (some participants were excluded subsequently from this analysis because of several erroneous diary entries about unrealistic amounts of food, e.g., 30,000 g chicken for lunch). Detailed statistics in [Tab. S6](#).

2.7. Relation between weight change and other meal timing-related factors

Besides increasing regularity, restricting meals to certain times of the day can also shorten the daily window in which meals are eaten and shift meals to other times of the day. In fact, our intervention caused many EG participants to avoid occasional meals at extremely early or late times, limiting their meals to much shorter intervals of 11 h (95% CI [10.468, 11.520]) ([Fig. 3A](#), [B](#), [Tab. S3](#)), while those of the CG remained long with 13.7 h (95% CI [12.902, 14.562]). Additionally, compared with before and with the CG, the EG intervention resulted in the first meal of the day being delayed to a later time for many subjects (EG: 09:21 (95% CI [08:58, 09:45]) vs. CG: 08:03 (95% CI [07:32, 08:33])) and the last meal of the day being advanced to an earlier time (EG: 20:17 (95% CI [19:59, 20:35]) vs. CG: 21:31 (95% CI [21:05, 21:57])) ([Fig. 3C](#), [D](#), [Tab. S3](#)). However, weight changes of participants were not related to one of these changes in eating characteristics ([Fig. 3B-D](#), [Tab. S6](#)).

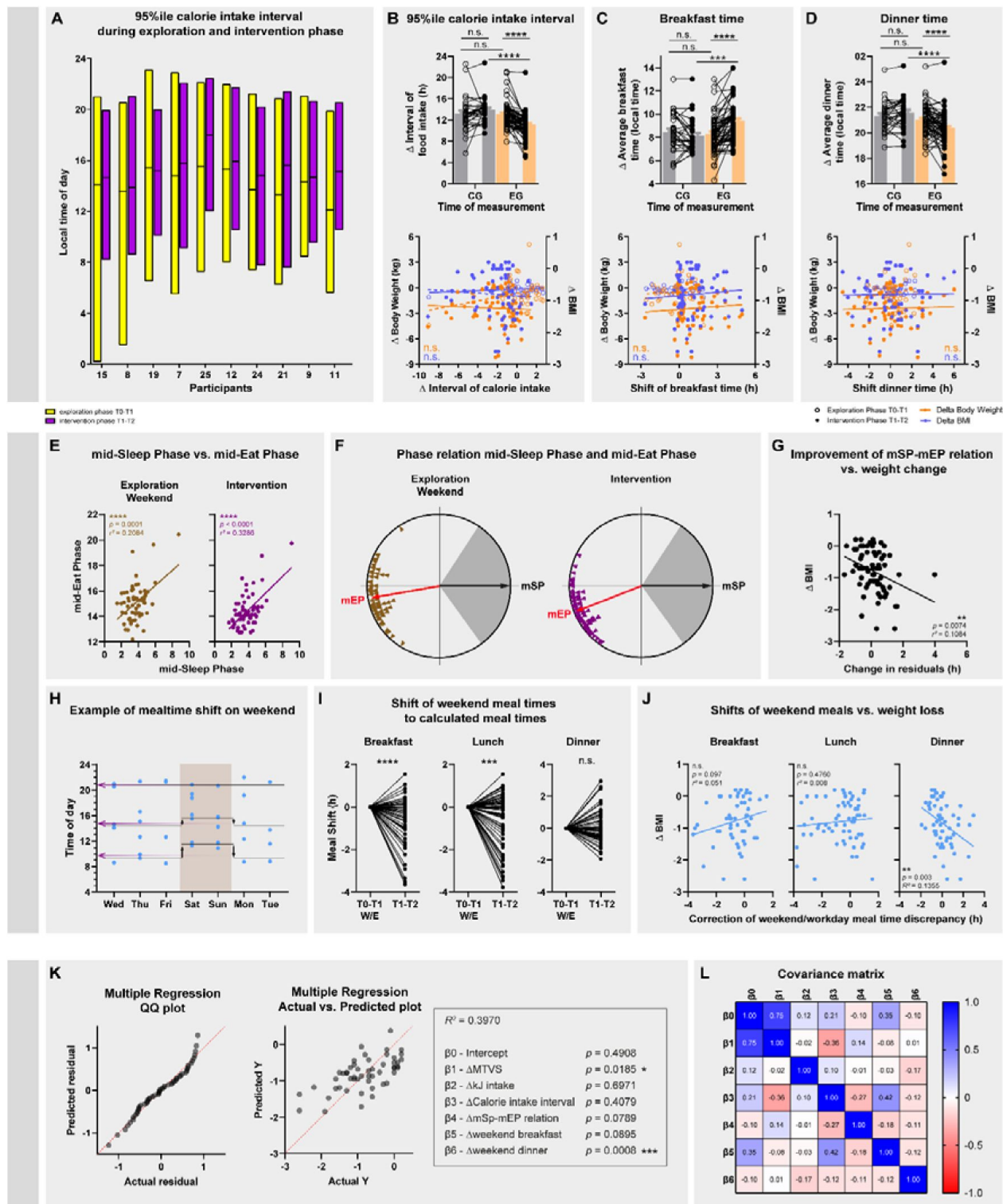


Fig. 3

Influence of other changes in eating times and of personalization of meal schedules on weight loss. (A) 95th percentile of the time period during which food was ingested by 15 example participants during the exploration phase (yellow) and the intervention phase (purple). (B) Change in length of this period and its relation to weight/BMI changes. (C, D) Change in time of first (C) and last (D) caloric intake and their association with weight/BMI changes. (B-D) 2-way repeated-measures ANOVA with Bonferroni post hoc test comparing CG and EG during the exploration or intervention phase and T0-T1 and T1-T2 within groups, *** $p \leq 0.001$, **** $p \leq 0.0001$. Linear regression. EG: $n=67$, varying n-numbers result from incomplete entries in the diaries. Detailed statistics in [Tables S3](#) and [S6](#). (E) The relation between mid-Sleep phase and mid-Eat phase (brown) on the weekends of the exploration phase. The goodness of fit of this association improves during through EG intervention and affects both workdays and weekend days (purple). Linear regression, $n=65$, **** $p \leq 0.0001$. Detailed statistics in [Table S6](#). (F) Relation of the mid-Sleep phase (mSP) and mid-Eat phase (mEP) during exploration and intervention phase. The circles represent 24 h, the gray area represents the estimated sleep time, and the black arrow represents the mid-sleep phase, which is normalized for all subjects. The colored triangles represent the mid-Eat phase of each subject in relation to their mid-Sleep phase. The red arrow shows the average mid-Eat phase across all subjects. $n=65$. Detailed statistics in [Table S10](#). (G) The smaller the residuals, i.e., the vertical distances of observation points from the regression line, became, the more BMI the EG subjects lost. Linear regression, $n=65$, ** $p \leq 0.01$. Detailed statistics in [Table S6](#). (H) Representative example of a meal diary showing a shift from meals on weekends. The purple arrows represent the calculated times, based on the events of the more frequent workdays, for the meal schedule of the intervention phase. (I) Shifts from weekend meals caused by the EG intervention. Paired t-test, $n=65$, *** $p \leq 0.001$, **** $p \leq 0.0001$. Data are normalized so that all baseline data at T0-T1 equal 0 to represent the change in hours. Detailed statistics in [Table S11](#). (J) Large phase shifts of breakfast and dinner, equivalent to a correction of the social jetlag of meals on weekends, are related to greater weight loss. Linear regression, breakfast: $n=55$, lunch: $n=64$, dinner: $n=63$, ** $p \leq 0.01$. Missing n-numbers are due to incomplete diary entries on weekends during the exploratory phase or because participants usually did not have a particular meal. Detailed statistics in [Table S6](#). (K) Results of a multiple regression to investigate the influence of individual variables while holding the others constant on the BMI of EG participants. The Quantile-Quantile (QQ) plot shows the data are normally distributed (left). The multiple regression plot and R^2 value shows that the multiple regression model explains the data well (middle). Additionally, the assignment of each regression beta coefficient to the independent variables is shown along with the p-values showing the influence on BMI. The data used are the same as those used in the individual regressions shown previously. Detailed statistics in [Tab. S9](#). (L) Covariance matrix with normalized covariances for each pair of parameters. Values close to 1 (blue) or -1 (red) indicate high intertwining of the data, values close to 0 (white) indicate independence or orthogonality of the data. The assignment of the coefficients to the independent variables is the same as in (K).

2.8. Effect of meal schedule personalization on weight loss

A key part of our hypothesis involves the personalization of meal schedules. Consistent with the concept of social jet lag, our data show that during the exploration phase, eating times on workdays differ from those on work-free days, and we expect eating times on non-workdays to correspond more closely to individual circadian characteristics. Indeed, our data show that the later the chronotype of participants (measured by the MCTQ mid-sleep phase, mSP [[24](#)]), the later their mid-eat phase (mEP) on weekends ([Fig. 3E](#), [Tab. S6](#)), with the mSP and mEP being almost antiphasic on average ([Fig. 3F](#), [Tab. S10](#)). The EG intervention maintains most of this relationship. However, the residuals of the regression become smaller, resulting in a model fitting the data better ([Fig. 3E](#), [Tab. S6](#)). Interestingly, the closer the intervention brought participants' observed values to their individual fitted values, i.e., the smaller the residuals became, the more weight they lost ([Fig. 3G](#), [Tab. S6](#)). From a clinical point of view, this indicates that a personalized adjustment of eating times to the individual chronotype, i.e., the establishment of a highly accurate phase relation between mEP and mSP, leads to a particularly effective reduction of excess weight.

The calculation of individual eating times was based on the frequency of hours at which meals were eaten during the exploration phase, which typically included more workdays than non-workdays (**Fig. 3H**). Consequently, for some subjects, the EG intervention created a discrepancy between what may have been physiologically optimal eating times, reflected by weekend data, and the calculated eating times (**Fig. 3I**, **Tab. S11**). However, the magnitude of this discrepancy has no effect on weight changes (**Fig. S20**, **Tab. S6**). On the contrary, if the calculated eating times resulted in large shifts of the weekend meals, this also means that previously particularly large “meal jet lags” between working and non-working days were eliminated. Indeed, the more the calculated eating times led to an advance of the first meal and a delay of the last meal, thus the greater the effect of the “meal jet lag” elimination was, the more the participants lost weight (**Fig. 3J**, **Tab. S6**). Interestingly, this also includes the elimination of too early dinners on work days. Our data show that a later dinner on all days of the week can lead to weight loss if this time corresponds to the calculated individual optimal time.

The independent variables used above for simple regressions may influence each other and therefore their influence on BMI may be confounded. Therefore, a multiple regression with the most important and least multicollinear variables (Δ MTVS, Δ kJ, Δ calorie intake interval, Δ mSP-mEP relation, Δ weekend breakfast, Δ weekend dinner) was done to study their individual influences on the BMI of EG participants while keeping the other variables constant. According to this analysis, only the change in MTVS and also the resulting elimination of “dinner jet lags” have significant influence on the BMI (**Fig. 3K**, **Tab. S9**). The elimination of “breakfast jet lags” and the improvement of the mSP-mEP ratio did not reach significance but showed a strong tendency to influence BMI. Changes in caloric intake or the time interval during which food is consumed had no effect on the BMI. A concurrent analysis of covariance shows that MTVS is most strongly intertwined with BMI, while other variables play a role to a much lesser extent (**Fig. 3L**).

2.9. General well-being outcomes

Since stable circadian rhythms are generally associated with many aspects of health, regular entrainment of a variety of peripheral and brain clocks by the *Zeitgeber* food may enhance overall well-being. To test this, participants were asked to complete clinical questionnaires measuring subjective health status, sleep quality, depressed mood, and self-efficacy at T0 and at T2. In addition, chronotype was also determined. Many aspects of subjective physical well-being (SF-36 - physical health items) improved significantly during the intervention in the EG (**Fig. 4A**, **Tab. S3**), reversing previous differences between CG and EG. In contrast, there was only a slight improvement in the general health item within the CG. In addition, sleep quality (PSQI) of the EG group improved significantly during the intervention, but not of the CG group, again eliminating prior differences between CG and EG. Furthermore, mental health, including vitality, depressed mood, and self-efficacy (SF-36 mental health items, IDS-SR, SWE) also improved significantly in EG participants, but not in the CG (**Fig. 4A**, **Tab. S3**). Consistent with our aim to optimize rather than to change individual rhythms, chronotype did not change during the study (**Fig. 4B**, **Tab. S3**).

3. Discussion

In our study, we show that increased regularity of the *Zeitgeber* food can lead to a significant reduction in body weight. Although the diary entries of some participants show that they changed their energy intake and also the food composition of fat, carbohydrates and proteins during the study, the observed weight loss is not related to these changes.

Related results have already been shown in mice, although here with a restriction to the entire active phase rather than specific eating times. Not surprisingly, mice gain weight and develop diabetes-like states under a high-fat diet [17]. It is surprising, however, that the negative

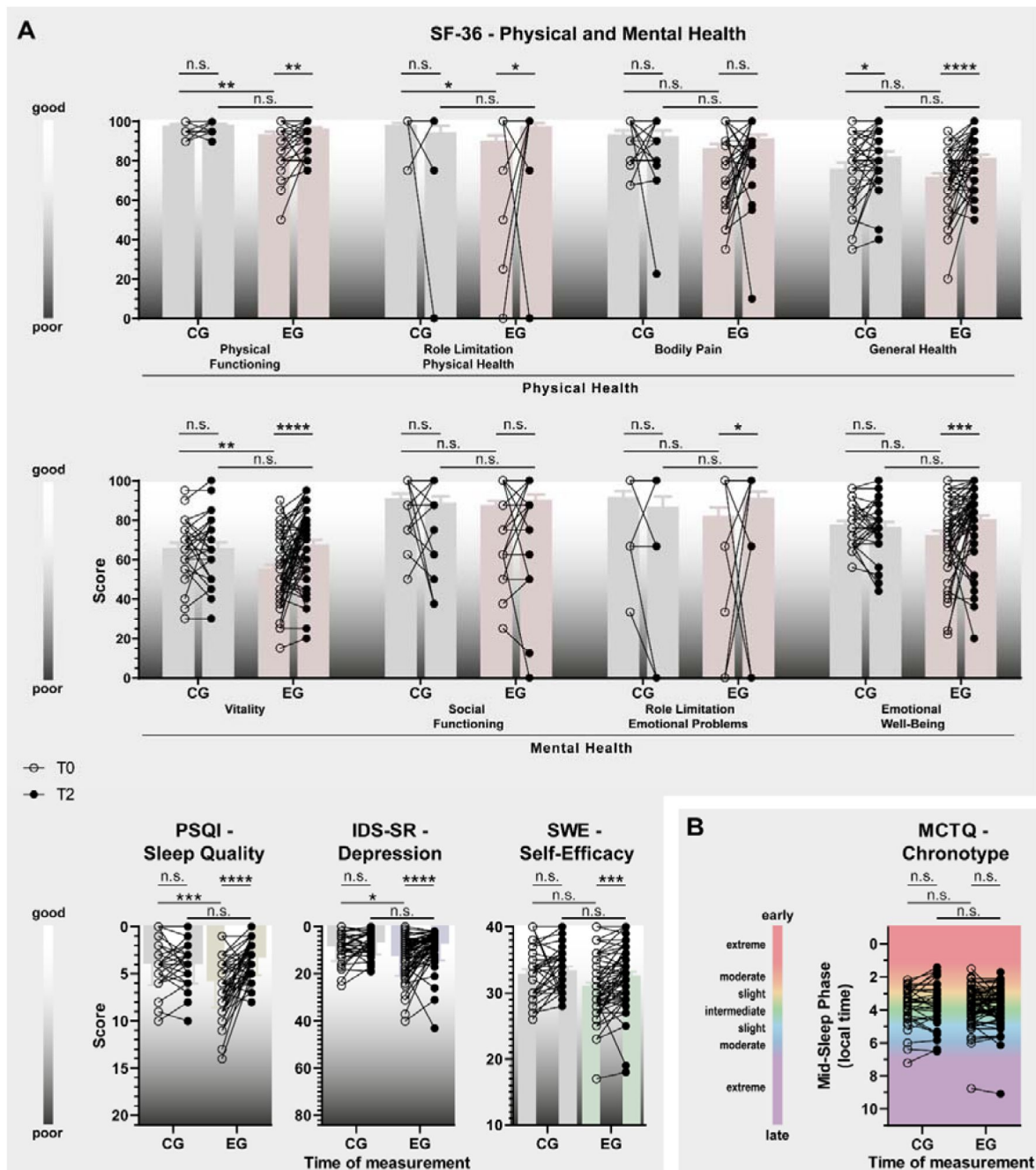


Fig. 4

Improvement in overall well-being over the course of the study. (A) Significant improvements are observed in various aspects of physical and mental health (SF-36), sleep quality (PSQI), depressed mood (IDS-SR), and self-efficacy (SWE) in participants of the EG (colored columns), but hardly among CG participants (gray columns). Scales are either upward or downward, depending on whether low or high scores on the questionnaires mean subjectively good or poor well-being. (B) The chronotype of EG and CG participants did not change during the study. 2-way repeated-measures ANOVA with Bonferroni post hoc test comparing CG and EG during the exploration or intervention phase and T0-T1 and T1-T2 within groups, * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$. CG: $n=33$; EG: $n=67$. Detailed statistics in [Tab. S3](#).

physiological consequences of the high-fat diet can be almost completely prevented if the mice consume the equivalent amounts of high-fat food exclusively during their activity phase and do not receive any food during their sleep phase [17, 25].

In our study, an increase in the regularity of lunch, and somewhat more strongly that of dinner, is related to weight loss, which to some extent contrasts with previously published data in which increased regularity of breakfast in particular, and not so much that of dinner, was identified as a predictor of weight loss [26]. Besides increased regularity, our intervention has additional effects on eating behavior, such as shortening the hours in which food is eaten, which in our study is not related to weight loss. This is similar to the principle of 16:8 intermittent fasting or time-restricted eating (TRE), in which meals are taken only specific time window, yet at unspecified times during that period. The effects of these protocols is mixed and often less pronounced, as less [21, 23, 27] weight loss or a longer period is required to achieve similar results [22] than in our study - with the caveat that more overweight and older participants participated in these studies than in ours. However, other studies have also explored the effect of restricting individual meals to specific times. In mice, restricting the same amount of high-fat chow to two times per day reduces the development of obesity [28]. And in humans, restricting meals to three fixed, uniform times leads to similar weight loss like we observed in our study - but in combination with calorie reduction and standardized food [29].

The fact that our participants lost weight to a similar extent despite free choice of meals could be due to the individual tailoring of meal schedules. It is true that our data show that scheduling single meals to supposedly circadian optimal eating times of the participants is not essential for successful weight loss. Circadian clocks can flexibly adjust to *Zeitgebers* and it is apparently more important to choose a time when individuals (have to) eat particularly frequently than the exact genetically determined eating time. However, our data still show that personalization of meal schedules is nevertheless useful. First, depending on the individual chronotype of the participants, their mealtime windows were earlier or later in the day. And those for whom the intervention optimized this relationship were particularly effective at losing excess weight. Second, if the schedule is designed to prevent large jumps between meals, especially dinner, on workdays and work-free days, weight loss is more successful.

Since food serves as a *Zeitgeber* for a large number of tissues, including several brain regions, it can be hypothesized that regular eating not only has a positive effect on body weight, but also on other aspects of well-being. A previous study by Panda et al. [18] has already shown that shortening the eating interval to 10-12 hours per day has a positive effect on the general well-being of the participants. Similarly, in our study, we also found significant improvements in different aspects of subjective well-being over the course of the study, specifically in physical and mental health, sleep quality, and self-efficacy. This shows that adhering to daily structured mealtimes can improve not only metabolic but also other physical and also psychological levels of health.

There are several explanatory approaches for why increased meal regularity can lead to weight loss despite equal caloric intake. For example, circadian clocks influence the composition of the microbiome, which regulates metabolic balance and body weight [30–34]. Additionally, circadian clocks help to anticipate mealtimes and provide the necessary digestive components in advance of food intake in advance [35–38]. Accordingly, the number of rhythmic genes in the liver of mice increases from ~350 to ~3,000 when the animals can follow their natural feeding rhythm and even to ~5,000 when the feeding time is restricted to a certain period of time [39].

3.1. Limitations

However, in our study, we did not collect metabolic parameters that could provide mechanistic explanations for the observed weight loss. Therefore, the assumption of an optimized metabolism remains speculation. Another limitation of the study is the self-assessment of key data on body

weight, eating times, and food composition. Body weight could not be collected in our clinic under supervision because of the Covid-19 pandemic. Regarding food intake, some participants underreported during the study, as indicated by unrealistically low kJ intakes. However, the individual reporting behavior of each participant remained largely the same during the exploration and intervention phases, so underreporting is not indicative of a change in eating behavior and, conversely, the observed kJ differences are likely due to actual changes in eating behavior. In addition, in the remaining evaluable 100 study participants, there was a significant difference in baseline body weight and BMI between CG and EG. However, even if CG participants had been heavier from the start, it cannot be assumed that the applied sham treatment with an eating window of 18 hours during wake time would have had a stronger effect. Second, those subjects in the EG group whose baseline BMI was similar to those of the CG subjects lost significant weight during the intervention, while CG participants did not. Taking these two arguments into account, it cannot be assumed that a floor effect, in which the CG participants cannot lose any more weight due to a lower baseline weight, is the reason for the lack of weight loss in the CG. Lastly, studies show that even much larger baseline BMI differences do hardly affect the success of weight loss programs [40, 41], so we assume that this also applies to our study. During the course of the study, there were some drop-outs in both groups. Importantly, these did not differ in baseline weight from the completers, which is why it can be ruled out that the intervention was not effective in a particular BMI group. Rather, the subjects who dropped out were those who either could not continue due to illness or did not adhere to the meal plan. However, the inclusion of drop-outs in the analysis of weight progression does not fundamentally change the results.

3.2. Conclusion

In summary, our data show that the increase of regularity of meals to a period of 90 min for each meal promotes a significant reduction of body weight and a significant increase of well-being within short time. Importantly, the successes achieved do not seem to be related to a reduction in calories, a change in food composition, or other eating behaviors. Additionally, we believe that this intervention poses hardly any health risks since it is most likely solely based on the optimization of metabolic processes. On the contrary, the strong increase in general well-being during the study rather indicates an improvement in health on many different levels.

4. Methods

4.1. Experimental approach

The study was a randomized controlled intervention trial with repeated measures investigating the effect of regularity of mealtimes on body weight and parameters related to overall well-being. Participants were blinded to the group assignment in that both control and experimental subjects were equally informed to participate in an intervention. The study was carried out at the Clinic for Psychiatry and Psychotherapy of the Ludwig Maximilian University (LMU), Munich, Germany. The study is registered at the German Clinical Trial Register with the trial number DRKS00021419 and was approved by the Ethics Committee of the LMU.

4.2. Participants

The participants were recruited across Germany via flyers at universities, fitness studios, adult education centers, pharmacies, grocery stores, and via Facebook® between September 2020 and August 2021. Those interested were able to contact the study staff by phone, email, or by making an entry in an online calendar tool. Participants had to be between 18-65 years old and have a BMI ≥ 22 kg/m². A requirement for participation in the study was that no other diet was concurrently followed, and that no medication was being taken regularly that could influence appetite or weight. No participants were included who knowingly suffer from a metabolic, mental, or

addictive disorder. Further exclusion criteria were pregnancy, blindness, bedriddenness, and dependence on assistance with eating. Special emphasis was placed on the participants' ability to understand the details of the study and to give written consent to participate.

4.3. Study Protocol

Each participant completed a 12-week program, which consisted of the following steps (Fig. S1C [\[18\]](#)):

4.3.1. Introductory Session and Initial Questionnaire Assessment

At the introductory session, the eligibility for possible participation in the study was determined and demographic data was collected. Then, suitable participants received a general introduction to the study, which included the theoretical background of the study and a detailed explanation of the study procedure. Subsequently, the participants were informed about their rights, the voluntary nature of their participation in the study, and data protection regulations. All participants had to sign an informed consent form for study participation and for the recording and use of their data. Participants weighed and measured their height themselves while study staff connected with them online via camera. Afterwards, the participant received questionnaires to fill out. This initial data assessment is referred to as T0 in the analysis.

4.3.2. Two-Week Exploration Phase

The introduction was followed by a two-week exploration phase in which the participants were asked to follow their usual eating habits and to document all caloric events (main meals, snacks, caloric drinks) using the smartphone application "FDDB" as accurately as possible. Based on this data, the personalized nutritional schedule was then developed. Data from the FDDB smartphone application also allowed us to calculate individual caloric intake along with the ratio of consumed macronutrients for each day during the study. Additionally, the participants weight was measured again. The end of the exploration phase is called T1 in the analysis.

4.3.3. Six-Week Intervention Phase

Experimental Group (EG): Following the exploration phase, the participants received their personalized mealtime schedule, which was prepared beforehand by the study staff based on the meal diary of the exploration phase. Together with the participants, the nutritional behavior of the last two weeks was reviewed and analyzed. Special attention was paid to irregularities in mealtimes and to times of day when food was eaten particularly frequently. In this way, participants were made aware that they eat irregularly, but that there are nevertheless times when their internal clock frequently triggers hunger. This was followed by a six-week intervention phase, in which the participants were instructed to have their meals only at times according to the mealtime schedule. Participants were explicitly told that they were not restricted in their choice and quantity of food. During the intervention phase, participants should continue to document all caloric events using the FDDB App. In addition, participants were asked to weigh themselves weekly to measure their progress. Halfway through the intervention phase, participants were contacted at least once by telephone to clarify possible questions and maintain compliance. Further, in case participants stopped making entries in their FDDB diary for two or more days, they were contacted to ask for reasons and to reinstate compliance again.

Control Group (CG): In principle, participants in the CG followed the same procedure. However, they were given a sham treatment, for which they were asked to restrict their meals to a freely chosen 18-hour time window per day, during which they could eat what they wanted and, above all, when they wanted. It can be assumed that restricting the eating time window to 18 hours should have no effect on weight, since such a long eating time window is hardly ever exceeded by most people anyway [\[18\]](#) and therefore hardly any change in eating behavior is to be expected.

4.3.4. Final Questionnaire Assessment

At the end of the intervention phase, the participants were asked to fill out the same questionnaires as at the introduction session. Additionally, weight was measured again. This data assessment is called T2 in the analysis.

4.3.5. Follow-up Assessment

Four weeks after completion of the study, the study staff contacted the participants again by telephone to inquire about their current weight and to obtain information about whether the participants had voluntarily continued the program after the end of the intervention phase. If the participants indicated that they did not continue the program, possible reasons were evaluated, and suggestions were received on how the current protocol could be improved to make it more attractive for permanent implementation in everyday life. Participants of the control group were informed at this point about their allocation and the actual question of the study. The final data assessment is called T3.

4.3.6. Restrictions due to the Covid-19 pandemic

Due to restrictions imposed by Covid-19, the study was carried out online. Data collection and communication was exclusively done via email, Zoom Meetings (Zoom Video Communications, Inc.), Facebook (Meta Platforms), or telephone.

4.4. Assessment of General Well-Being

To measure whether restricting meals to times when we believe the circadian system triggers hunger leads to an increase in general well-being, questionnaires were collected to measure sleep quality, physical well-being, self-efficacy, and depressive mood. Additionally, the chronotype was assessed (Tab. S12 [↗](#)).

4.5. Mealtime Diary with Fddb Application

The nutrition diary of the smartphone application Fddb was used to document all caloric events. Fddb is freely available in the usual application stores and is operated by the independent food industry company Fddb Internetportale GmbH, Berlin. Users can record each caloric event with detailed information about the amount, type, and method of preparation of each food, which can be selected from a large database of different foods, which also contains data about the energy value of each product. The app automatically records the date and time of each caloric event. In case a certain food is not listed in the database, placeholder events with estimated energy values can be recorded. In addition to the participants, the study staff also had access to each account to check compliance and regularly download diary data.

4.6. Creation of Personal Meal Schedules

At the end of the exploration phase, Fddb data were downloaded, and clusters of times of meals were identified. First, for each participant the optimal number of mealtime clusters was determined by the heuristic technique *elbow method*. With the number of meals calculated in this way, the optimal times for each of these meals were then calculated using the *k-means* algorithm with the scikit-learn machine learning package for Python. The variable minutes was created to adjust the format of the clock times in the data. The *k-means* algorithm finds a clustering structure, that minimizes the sum squared error which measures the distance of each datapoint to its representative value. The center of each cluster was then calculated as the mean of all observations in the cluster, which was extracted in hours and minutes. Calculated times were rounded to quarter, half, three-quarter, or whole hours and visually displayed in scatterplots for discussion with participants of the EG. If the calculated times were incompatible with, for

example, participants' work schedules, they had the opportunity to adjust the times of meals accordingly. Then, an agreement was reached to eat only at these times for the next six weeks. For each meal, a time window of plus/minus 30 minutes around the specified time was granted. The participants were instructed not to advance missed meals or to make up for them at times that did not correspond to the established times, but to wait for the next established mealtime if possible. A scatterplot was also created for participants of the CG, but they were not provided with meal timings; instead, an 18-hour time window for meals was established with them.

4.7. Statistical Analyses

Statistical analyses were performed with SPSS 24, Python, GraphPad Prism 9.2.0, and R. Details about statistical tests used for specific experiments are indicated in the main text and the corresponding figure legends.

4.7.1. Mealtime Variability Score (MTVS)

A Python script was written to generate the MTVS, which is calculated from the deviation of each caloric event from the specified mealtime closest to the event. Similar to a previous study, each meal was given a score value based on the time deviation of the specified meal in minutes: 1 = +/- 0-15 min, 2 = +/- 16-30 min, 3 = +/- 31-45 min, 4 = +/- 46-60 min, 5 = +/- 61-75 min, 6 = +/- 76-90 min, 7 = +/- 91-105 min, 8 = +/- 106-120 min, 9 = +/- 2-3 h, 10 = +/- 3-4 h, 11 = over +/- 4 h [20]. Since the participants were given a 60-minute time slot for each meal, a score of up to 2 is considered very regular. Based on individual score values of each meal, an average score value can be calculated for each individual day, for each individual week or for the entire period, i.e., a daily, a weekly, or a total score. Similarly, daily, weekly, and total scores can be calculated for each type of meal, i.e., breakfast, lunch, dinner.

4.7.2. Sample Size and randomization

For the sample size calculation, we assumed an average of 2.5 kg of body weight loss, based on TRE studies [18, 19, 21–23], as we are not aware of any other study in which meal times were personalized and scheduled. The reported standard deviation for body weight loss in such studies is 3.2 [19]. Randomization was to be 2:1 (EG:CG) because no significant change in body weight was expected from the control treatment. The randomized assignment was carried out using a continuous list prepared by the study supervisor with the Excel® randomization function. Participant enrollment and assignment was carried out by study staff. On this basis a large effect size $d = 0.8$ was assumed for the calculation of the power analysis in G*Power and transformed into the effect size $f = 0.4$. The sample size calculation was performed a priori for two groups (EG and CG) and the main three measuring time points (T0, T1, T2) resulting in a sample size of $N = 100$ participants.

4.7.3. Within- and Between Group Comparisons across T0-T1 and T1-T2

Two-way repeated-measures ANOVA was used to compare group and time effects in the CG and EG and in the exploration (T0-T1) and intervention (T1-T2) phases, as well as effects of the interaction of group and time. Additionally, Bonferroni's multiple comparison tests were used to compare effects of time within groups (T0-T1 vs. T1-T2 within CG or EG) and effects between groups at either the exploration or intervention phase (CG vs. EG at T0-T1 or T1-T2). This form of analysis was applied to data in **Figures 1C**, **2A & E**, **3B-D**, **4A & B**, and **S2J**. If missing data points prevented a repeated-measure analysis, a mixed-effects model with Bonferroni's multiple comparison test was applied instead. This applies to the data in **Figures 1F**, **S2C, G & M**. When comparisons were made only between exploration and intervention phases within a group, a paired t-test was applied, which is the case for data shown in **Figure 3I**. When comparisons were made with groups with very different n-numbers, Welch's test was used instead of the t-test. This applied to data shown in **Figure S2A**.

4.7.4. Body Weight Development over Time

To analyze the change in body weight within the CG and EG groups over the T0-T2 period, one-way repeated measures ANOVA with Bonferroni's multiple comparison test was applied. The division of CG and EG into continuing and discontinuing did not allow inclusion of T3 data in the repeated measures analysis. Therefore, the comparison between T2 and T3 was completed with a paired t-test. This type of analysis was applied in data of **Figures 1E** and S2A.

4.7.5. Relationships between parameters

Simple linear regression was used to determine whether two different parameters were related. This analysis was performed with data from **Figures 1G, 2C, D & F, 3B-E, G & J, and S2H, I, L, N & O**. Relationships of more than two parameters were analyzed with multiple regression, which applies to data from **Figures 1H** and **3K**.

4.7.6. Circadian Phase Distribution

The mSP was evaluated according to the MCTQ [24]. The mEP was calculated in the same way, but instead of using the time of falling asleep and the time of waking up as in the calculation of the mSP, the first and the last meal of the day were used for the calculation. The distribution of each calculated phase was analyzed using Rayleigh's uniformity test. This analysis concerns the data shown in **Figure 3F**.

Abbreviations

- CG: Control group
- EG: Experimental group
- IDS-SR: Self-Assessment Inventory of Depressive Symptoms
- MCTQ: Munich Chronotype Questionnaire
- MTVS: Meal time variability score
- PSQI: Pittsburgh Sleep Quality Index
- SF-36: 36-Item Short Form Health Survey
- SWE: Scale of General Expectations of Self-Efficacy

Data availability

All data produced in the present study are available upon reasonable request to the authors.

Code availability

The codes used to calculate MTVS, individual optimal number of meals, and personalized optimal eating times will be available at <https://github.com/dolandgraf/Time-To-Eat.git> as of the date of publication of this paper.

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Author Contributions

DL, AHL, IW: Conceptualization

IW, JT, JMO, TJGB, DL: Data curation

DL: Formal Analysis

DL: Funding acquisition

IW, JT, JMO, TJGB: Investigation

IW, DL, AHL: Methodology

DL: Project administration

DL, IW: Resources

IW, JT: Software

DL: Supervision

DL, IW, AHL: Validation

DL: Visualization

DL: Writing – original draft

DL, AHL, IW: Writing review & editing

Ethics declaration

The conduct of the study was evaluated and approved by the Ethics Committee of Ludwig Maximilian University under file number 19-975.

Competing interests

The authors declare no competing interest.

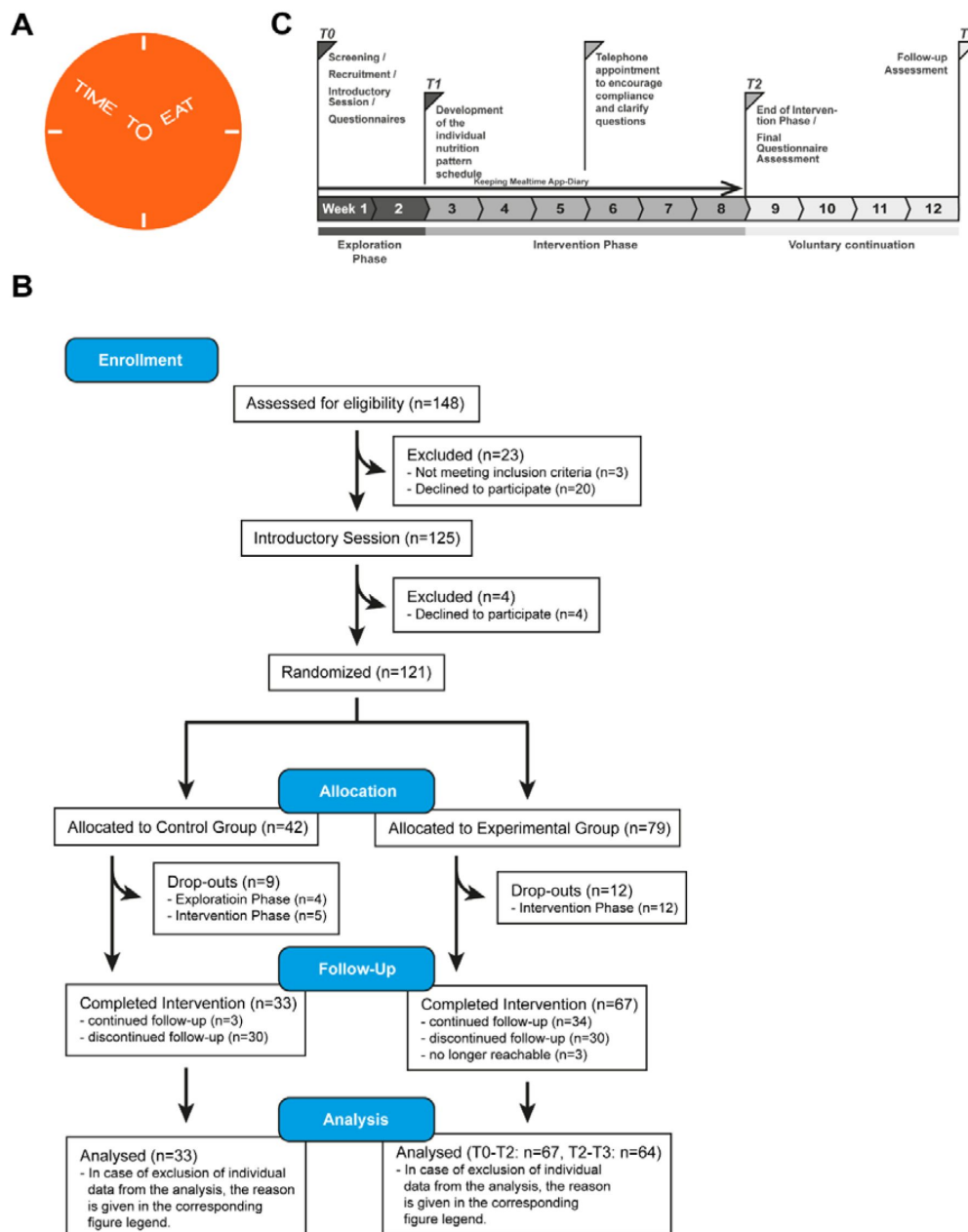


Fig. S1

(A) Logo of the “Time To Eat” study. The logo represents a clock in which the letters act as hands. (B) Consort diagram through the phases of the trial. A total of 148 initial participants were recruited. Of these, 48 were excluded or withdrew from the study in the course of the trial. Thus, the final study sample consisted of 100 participants of whom 67 were allocated to the EG and 33 to the CG. Until the follow-up appointment, 34 EG participants voluntarily continued the intervention. 30 EG participants discontinued the intervention but were available for further data collection, while three EG participants were no longer available and thus no analyzable data were available from them at T3. Of the 33 CG participants, only three voluntarily continued the sham intervention until T3. (C) Timeline of the study. **INTRODUCTORY SESSION:** Test participants are screened, recruited by a member of the study staff, and sign the consent form. This is followed by an introductory session where they are informed about the contents of the study and familiarized with the FDDB smartphone application. In addition, questionnaires are filled out and weight and abdominal girth are measured. **EXPLORATION PHASE:** During the following two weeks, the participants are asked to record each caloric event with the FDDB app. Afterwards, a mealtime schedule is developed for each participant. **INTERVENTION PHASE:** During the following six weeks, the participants are instructed to take their meals only at times according to the mealtime schedule and to report their weight weekly. In the middle of the intervention phase, a telephone appointment is made to maintain compliance. **FINAL QUESTIONNAIRE ASSESSMENT:** At the end of the intervention phase, the participants are asked to fill out the same questionnaires as at the introduction session. **FOLLOW-UP ASSESSMENT:** After another four weeks, the test participants are contacted again to explore whether they have continued the intervention voluntarily and to ask again for their body weight.

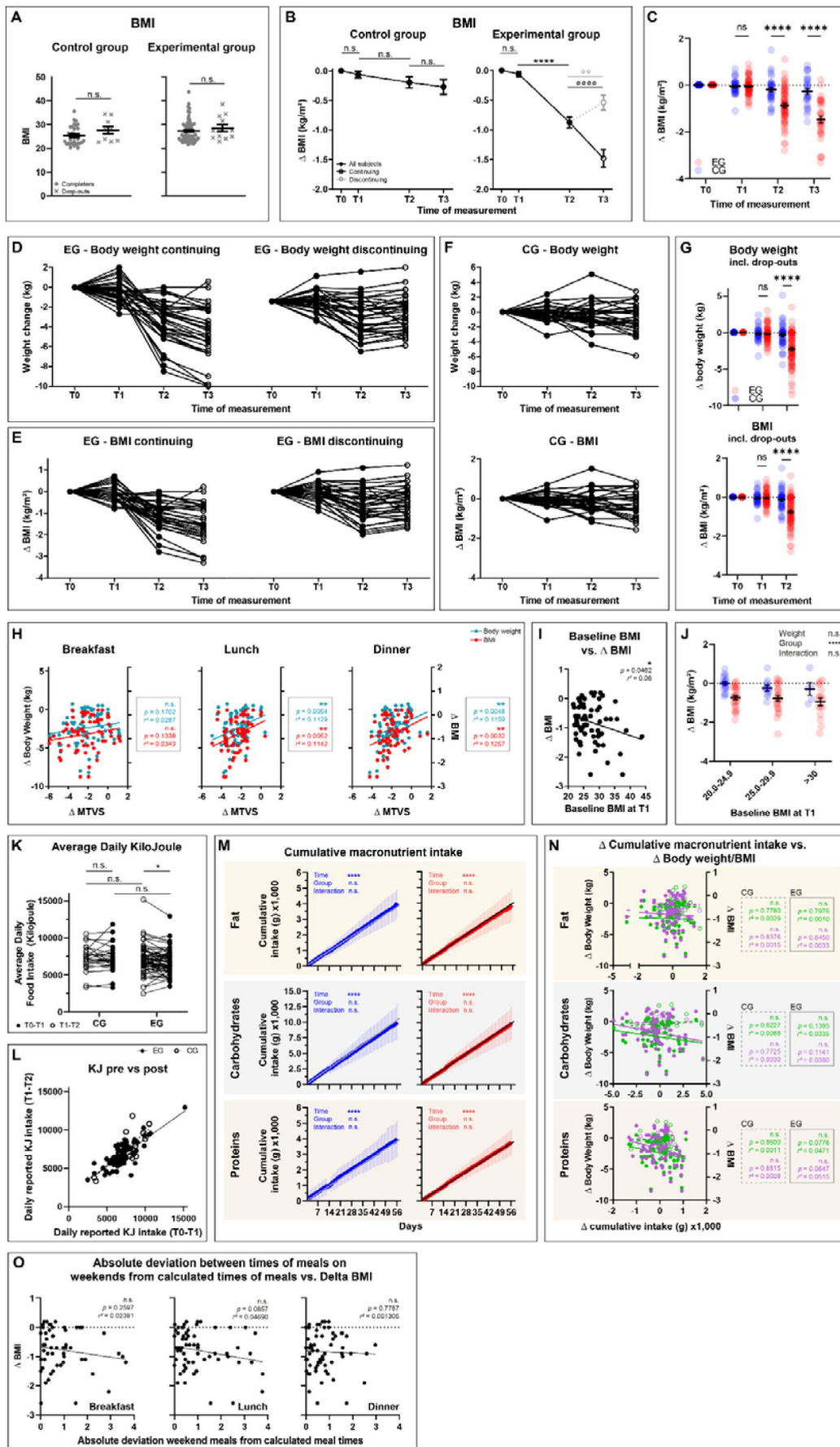


Fig. S2

(A) BMI differences between study completers and drop-outs. Differences were calculated with a Welch's test (detailed statistics in [Tab. S2](#)). (B) Within group differences of BMI during the study. BMI was collected at different time points, at the beginning of the study (T0), after the exploration phase (T1), after the intervention phase (T2), and after the follow-up phase (T3). During the follow-up phase, 34 EG participants voluntarily continued the intervention (continuing; black line and open circle) while 33 participants dropped out (discontinuing; gray, dashed line and open circle). Only three control subjects continued their intervention in the follow-up phase, which is why a division into continuing and discontinuing was omitted in this group. Data are normalized so that all baseline data at T0 equals 0 to represent change. Differences of all CG and EG participants between T0, T1, and T2 were calculated with a 1-way ANOVA, as were differences between T2 and T3 of CG participants (Bonferroni post-hoc test, **** $p \leq 0.0001$). Differences between T2 and T3 of continuing and discontinuing EG participants were calculated with a paired t-test $p \leq 0.01$, **** $p \leq 0.0001$ (detailed statistics in [Tab. S4](#)). (C) Between group differences of BMI from T0-T3. Mixed-effect model with Bonferroni post-hoc test comparing CG and EG at different phases during the study, **** $p \leq 0.0001$ (detailed statistics in [Tab. S5](#)). (D) Individual body weight (E) and BMI (F) changes of all 34 continuing and all 33 discontinuing participants of the EG group and of CG participants (F) during the study. Data is normalized so that all baseline data at T0 equals 0 to represent change. (G) Between group differences of body weight and BMI from T0-T3 including the last available values of study drop-outs which were carried forward. Mixed-effect model with Bonferroni post-hoc test comparing CG and EG at different phases during the study, **** $p \leq 0.0001$ (detailed statistics in [Tab. S5](#)). (H) Significant correlation between body weight/BMI and MTVS of lunch and dinner. The more participants improved their regularity of lunch and dinner (decreasing MTVS), the more weight/BMI they lost. Linear regression, CG: $n=33$; EG: $n=67$, ** $p \leq 0.01$ (detailed statistics in [Tab. S6](#)). (I) Correlation between baseline BMI at T1 and change in BMI in EG participants. Linear regression, $n=67$ (detailed statistics in [Tab. S6](#)). (J) Between group differences after stratification according to different baseline BMIs. 2-way repeated-measure ANOVA with Bonferroni post-hoc test, *** $p \leq 0.001$ (detailed statistics in [Tab. S8](#)). (K) Raw data of change in caloric intake during intervention. 2-way repeated-measure ANOVA with Bonferroni post-hoc test, CG: $n=30$; EG: $n=67$, * $p \leq 0.05$, (detailed statistics in [Tab. S3](#)). (L) Correlation between reported ingested energy during the exploration phase vs. the intervention phase. Linear regression, CG: $n=30$; EG: $n=67$ (some participants were excluded from this analysis because of several erroneous diary entries about unrealistic amounts of food), (detailed statistics in [Tab. S6](#)). (M) Determined and estimated cumulative intake of fat, carbohydrates, and proteins during the study. Mixed-effects model with Bonferroni post hoc test comparing determined and estimated cumulative energy intake. The estimated macronutrient intake is formed as an assumed continuation of the average daily intake during the exploration phase. The Bonferroni post hoc test showed no significant differences on any study day. (N) No significant correlation between body weight/BMI and changes of cumulative macronutrient intake. Linear regression, CG: $n=30$; EG: $n=67$, (detailed statistics in [Tab. S6](#)). (K, L, M, N) Some participants were excluded from this analysis because of several erroneous diary entries about unrealistic amounts of food. (O) The absolute deviation between the times of weekend meals and the times of food intake calculated for the EG intervention did not affect the subjects' BMI change. Linear regression, $n=67$ (detailed statistics in [Tab. S6](#)).

Characteristic	CG (n = 33; male: 1. female: 23)			EG (n = 67; male: 26, female: 41)			Difference CG - EG	
	Mean	Range	SD	Mean	Range	SD	p-value	Summary
BMI (kg/m ²) incl. drop-outs	25.9	20.3 / 35.7	4.138	27.5	21.7 / 43.8	4.627	.0597	n.s.
Age (years)	33	20-60	12.08	38	19 - 64	14.49	.0795	n.s.
BMI (kg/m ²) (T0/T1)	25.5 / 25.4	20.4 – 35.8 / 20.3 – 35.7	3.83 / 3.88	27.4 / 27.3	22.0 – 43.6 / 21.7 – 43.8	4.62 / 4.59	.0420 / .0420	* / *
BMI (T2)	25.3	20.4 – 34.6	3.77	26.5	21.2 – 42.5	4.47	.1680	n.s.
Δ BMI (T0-T2)	-.20	-1.17 – 1.52	.55	-.87	-2.8 – 1.0	.72	.0420	*
Body weight (kg) (T0/T1)	75.04 / 74.87	54.2 – 134.6 / 54.0 – 134.2	16.94 / 17.07	82.73 / 82.52	58.5 – 129.0 / 58.1 – 129.5	16.72 / 16.59	.0338 / .0342	* / *
Body weight (T2)	74.48	53.5 – 130.2	16.84	80.11	56.3 – 125.8	16.30	.1116	n.s.
Δ Body weight (T0-T2)	-.56	-4.4 – 5.1	1.72	-2.62	-8.5 – 3.5	2.23	.0338	*
MTVS exploration (total)	3.35	1.32 – 4.79	.8081	3.88	1.11 – 5.76	.9086	.0058	**
MTVS intervention (total)	3.94	2.45 – 5.13	.7203	1.72	1.02 – 3.84	.6156	<.0001	****
ΔMTVS (total)	.583	-.940 – 3.47	.9051	-2.16	-4.69 - .420	.9571	<.0001	****
KJ intake exploration	7,211	3,350 – 1,537	1,596	7,071	2,505 – 15,185	1,854	.7197	n.s.
KJ intake intervention	7,248	3,324 – 11,829	1,766	6,723	3,496 – 12,932	1,661	.1618	n.s.
% Fat exploration	20.73	11.59 – 26.12	3.639	21.85	10.86 – 32.30	3.811	.1773	n.s.
% Fat intervention	22.06	12.54 – 28.98	3.282	21.40	11.70 – 28.91	3.158	.3477	n.s.
% Carbohydrate exploration	57.51	47.63 – 71.40	6.536	56.23	42.69 – 71.11	6.247	.3607	n.s.
% Carbohydrate intervention	56.54	45.29 – 70.90	5.712	55.90	35.20 – 68.59	5.282	.5969	n.s.
% Protein exploration	21.77	12.56 – 31.56	4.456	21.92	13.29 – 35.51	4.520	.8747	n.s.
% Protein intervention	21.41	15.07 – 30.13	3.688	22.70	16.80 - 39.78	40.70	.1398	n.s.

Table S1

Baseline characteristics of final 100 participants.

Fig.	Group	Variable value (mean ± SD)				Statistical test	d.f.	t-value	Significance	
		Completers	n-value	Drop-outs	n-value				p-value	Summary
S2A	CG	25.40 ± 3.882	33	27.65 ± 4.793	9	Welch's test	11.03	1.298	.2207	n.s.
	EG	27.32 ± 4.586	67	28.53 ± 4.932	12				.4437	n.s.

SD = Standard deviation

d.f. = degree of freedom

n.s. = not significant

Table S2

Statistical data on the difference between completers and study drop-outs. The data in the table refers to the data shown in Fig. S2A [↗](#).

Fig.	Meal	MTVS (mean; 95% CI)						Statistical test	ANOVA results	Source of Variation			Bonferroni post-hoc								
		CG			EG								Within Group (T0-T1 vs. T1-T2)				Between Group (CG vs. EG)				
		Exploration phase	Intervention phase	n-value	Exploration phase	Intervention phase	n-value			p-val.	Sum.	p-val.	Sum.	T0-T1 p-val.	Sum.	T1-T2 p-val.	Sum.				
1C	Breakfast	3.257; (2.742, 3.773)	4.352; (3.755, 4.948)	33	3.893; (3.508, 4.278)	1.722; (1.543, 1.902)	66	Two-way RM ANOVA	d.f.	1, 97	1, 97	1, 97	.0012	**	<.0001	****	.0564	n.s.	<.0001	****	
	Lunch	3.331; (2.93, 3.731)	3.743; (3.352, 4.135)	33	4.021; (3.75, 4.292)	1.668; (1.484, 1.852)	67		F-value	21.20	8.111	74.63									
									p-value	<.0001	.0054	<.0001									
									Summary	****	**	****									
									d.f.	1, 98	1, 98	1, 98	.0543	n.s.	<.0001	****	.0030	**	<.0001	****	
	F-value	14.45	74.51	151.4																	
p-value	.0003	<.0001	<.0001																		
Summary	***	****	****																		
Dinner	3.358; (3.033, 3.633)	3.807; (3.527, 4.087)	33	3.740; (3.477, 4.002)	1.778; (1.619, 1.937)	64	d.f.	1, 95	1, 95	1, 95	.0384	*	<.0001	****	.0814	n.s.	<.0001	****			
							F-value	32.57	42.20	107.8											
							p-value	<.0001	<.0001	<.0001											
							Summary	****	****	****											
1D	Total	3.353; (3.066, 3.640)	3.936; (3.681, 4.192)	33	3.879; (3.658, 4.101)	1.723; (1.573, 1.873)	67	Two-way RM ANOVA	d.f.	1, 98	1, 98	1, 98	.0011	**	<.0001	****	.0032	**	<.0001	****	
							F-value		41.87	61.87	187.7										
							p-value		<.0001	<.0001	<.0001										
							Summary		****	****	****										
2A	Daily KJ intake (x-fold change)	1.0; (... .0)	1.011; (.9599, 1.062)	30	1.0; (... .0)	.9718; (.927, 1.017)	67	Two-way RM ANOVA	d.f.	1, 95	1, 95	1, 95	>.9999	n.s.	.3604	n.s.	>.9999	n.s.	.2877	n.s.	
							F-value		.2152	1.077	1.077										
							p-value		.6438	.3020	.3020										
							Summary		n.s.	n.s.	n.s.										
	S2K	Average daily KJ intake	7212; (6615, 7808)	7248; (6588, 7907)	30	7071; (6619, 7523)	6723; (6318, 7128)		67	Two-way RM ANOVA	d.f.	1, 95	1, 95	1, 95	>.9999	n.s.	.0271	*	>.9999	n.s.	.3416
							F-value	1.571	.8500		.1256										
							p-value	.2131	.3589		.1256										
							Summary	n.s.	n.s.		n.s.										
2C		% Fat	20.73; (19.37, 22.09)	22.06; (20.83, 23.28)	30	21.85; (20.92, 22.78)	21.40; (20.63, 22.17)	67	Two-way RM ANOVA		d.f.	1, 95	1, 95	1, 95	.0394	*	.4577	n.s.	.2895	n.s.	.7767
							F-value	1.686		.1115	6.998										
							p-value	.1973		.7391	.0095										
							Summary	n.s.		n.s.	**										
	% Carbo-hydrates	57.51; (55.07, 59.95)	56.54; (54.4, 58.67)	30	56.23; (54.7, 57.75)	55.90; (54.62, 57.19)	67	d.f.		1, 95	1, 95	1, 95	.4796	n.s.	>.9999	n.s.	.6494	n.s.	>.9999	n.s.	
							F-value	1.719	.6365	.4293											
							p-value	.1930	.4270	.5139											
							Summary	n.s.	n.s.	n.s.											
% Pro-teins	21.77; (20.1, 23.43)	21.41; (20.03, 22.78)	30	21.82; (20.82, 23.02)	22.70; (21.71, 23.69)	67	d.f.	1, 95	1, 95	1, 95	.9850	n.s.	.0564	n.s.	>.9999	n.s.	.3322	n.s.			
							F-value	.4446	.6848	3.284											
							p-value	.5065	.4100	.0731											
							Summary	n.s.	n.s.	n.s.											
3B	95%ile calorie intake interval	13.29; (12.24, 14.34)	13.73; (12.9, 14.56)	32	13.00; (12.45, 13.56)	10.99; (10.47, 11.52)	67	Two-way RM ANOVA	d.f.	1, 97	1, 97	1, 97	0.4686	n.s.	<.0001	****	>.9999	n.s.	<.0001	****	
							F-value		12.31	11.15	30.00										
							p-value		.0007	.0012	<.0001										
							Summary		***	**	****										
	3C	Breakfast time	8.39 ± 1.685	8.06 ± 1.434	33	8.24 ± 1.516	9.37 ± 1.633		67	Two-way RM ANOVA	d.f.	1, 98	1, 98	1, 98	0.2939	n.s.	<.0001	****	>.9999	n.s.	.0002
							F-value	8.122	3.637		27.49										
							p-value	.0053	.0594		<.0001										
							Summary	**	n.s.		****										
3D		Dinner time	21.26 ± 1.139	21.53 ± 1.217	32	20.96 ± 1.143	20.3 ± .6156	67	Two-way RM ANOVA		d.f.	1, 97	1, 97	1, 97	0.2969	n.s.	<.0001	****	.4725	n.s.	<.0001
							F-value	3.199		11.06	17.51										
							p-value	.0768		.0012	<.0001										
							Summary	n.s.		**	****										

Table S3

Statistical data from all Two-way-RM ANOVAs analyzing time, group, as well as interaction effects between time x group. In data for **Figure 1**, missing p-values in the and EG are due to the different number of meals for each participant. In the data for **Figure 2**, some participants were excluded from this analysis because of several probably erroneous diary entries about unrealistic amounts of food. In data for **Figure 3**, varying n-numbers result from incomplete entries in the diaries. In data for **Figure 4**, missing data are due to incomplete questionnaires.

4A	SF-36: PF	98.03; (96.95, 99.11)	98.33; (97.38, 99.29)	33	93.73; (91.45, 96.01)	96.27; (94.79, 97.75)	67	Two-way RM ANOVA	d.f.	1, 98	1, 98	1, 98	>.9999	n.s.	.0020	**	.0055	**	.2940	n.s.
									F-value	4.769	6.375	2.951								
									p-value	.0314	.0132	.0890								
									Summary	*	*	n.s.								
	SF-36: RLPH	98.49; (96.34, 100.63)	94.70; (88.14, 101.26)	33	90.30; (84.89, 95.71)	97.76; (94.63, 100.90)	67		d.f.	1, 98	1, 98	1, 98	0.7540	n.s.	.0288	*	.0464	*	.7856	n.s.
									F-value	.4966	1.092	4.655								
									p-value	.4827	.2985	.0334								
									Summary	n.s.	n.s.	*								
	SF-36: BP	93.64; (89.86, 97.42)	92.73; (87.2, 98.25)	33	86.57; (82.22, 90.91)	91.57; (88.05, 95.08)	67		d.f.	1, 98	1, 98	1, 98	>.9999	n.s.	.0779	n.s.	.0626	n.s.	>.9999	n.s.
									F-value	.9674	2.688	2.017								
							p-value	.3278	.1043	.1586										
							Summary	n.s.	n.s.	n.s.										
SF-36: GH	76.06; (69.58, 82.54)	82.27; (77.05, 87.50)	33	72.01; (68.64, 75.39)	81.79; (78.78, 84.81)	67	d.f.	1, 98	1, 98	1, 98	0.0263	*	<.0001	****	.3732	n.s.	>.9999	n.s.		
							F-value	28.30	.7258	1.406										
							p-value	<.0001	.3963	.2386										
							Summary	****	n.s.	n.s.										
SF-36: Vit	66.06; (61.07, 71.05)	65.91; (60.16, 71.66)	33	55.97; (51.97, 59.97)	68.62; (64.37, 72.87)	67	d.f.	1, 98	1, 98	1, 98	>.9999	n.s.	<.0001	****	.0084	**	.8745	n.s.		
							F-value	11.66	1.552	12.23										
							p-value	.0009	.2158	.0007										
							Summary	***	n.s.	***										
SF-36: SF	91.29; (86.65, 95.92)	89.02; (82.48, 95.55)	33	87.68; (83.233, 92.14)	90.63; (85.97, 95.28)	67	d.f.	1, 98	1, 98	1, 98	0.9033	n.s.	.3271	n.s.	.6901	n.s.	>.9999	n.s.		
							F-value	.03326	.08925	2.023										
							p-value	.8557	.7658	.1581										
							Summary	n.s.	n.s.	n.s.										
SF-36: RLEP	91.92; (85.99, 97.85)	86.87; (76.24, 97.50)	33	83.58; (75.41, 91.76)	93.04; (87.64, 98.43)	67	d.f.	1, 98	1, 98	1, 98	0.7519	n.s.	.0393	*	.2999	n.s.	.5728	n.s.		
							F-value	.4026	.05550	4.371										
							p-value	.5272	.8143	.0392										
							Summary	n.s.	n.s.	*										
SF-36: EWB	77.94; (74.36, 81.52)	76.85; (72.24, 81.46)	33	72.93; (68.74, 77.11)	81.07; (77.08, 85.07)	67	d.f.	1, 98	1, 98	1, 98	>.9999	n.s.	.0002	***	.2491	n.s.	.3901	n.s.		
							F-value	4.207	.02041	7.210										
							p-value	.0429	.8867	.0085										
							Summary	*	n.s.	**										
PSQI	4.032; (3.177, 4.887)	4.097; (3.325, 4.868)	31	5.921; (5.219, 6.622)	3.365; (2.894, 3.836)	63	d.f.	1, 92	1, 92	1, 92	>.9999	n.s.	<.0001	****	.0006	***	.3055	n.s.		
							F-value	21.75	1.776	24.06										
							p-value	<.0001	.1860	<.0001										
							Summary	****	n.s.	****										
IDS-SR	8.818; (6.621, 11.016)	7.121; (5.312, 8.931)	33	12.94; (10.86, 15.02)	7.662; (5.887, 9.436)	66	d.f.	1, 96	1, 96	1, 96	0.3229	n.s.	<.0001	****	.0155	*	>.9999	n.s.		
							F-value	22.30	3.019	5.877										
							p-value	<.0001	.0855	.0172										
							Summary	****	n.s.	*										
SWE	32.70; (31.28, 34.12)	33.21; (31.99, 34.43)	33	30.87; (29.79, 31.94)	32.46; (31.36, 33.57)	67	d.f.	1, 98	1, 98	1, 98	.7093	n.s.	.0002	***	.0873	n.s.	.8140	n.s.		
							F-value	9.741	2.382	2.556										
							p-value	.0024	.1259	.1131										
							Summary	**	n.s.	n.s.										
4B	MCTQ	3.802; (3.37, 4.234)	3.720; (3.223, 4.217)	32	3.746; (3.465, 4.026)	3.828; (3.542, 4.114)	66	d.f.	1, 98	1, 98	1, 98	>.9999	n.s.	.8189	n.s.	>.9999	n.s.	>.9999	n.s.	
							F-value	1.85e ⁻⁶	.01149	.8940										
							p-value	.9989	.9149	.3468										
							Summary	n.s.	n.s.	n.s.										

SD = Standard deviation
RM = Repeated Measure
n.s. = not significant
PF = Physical functioning
BP = Bodily pain
VT = Vitality
RE = Role of limitation emotional Problems
PSQI = Pittsburgh Sleep Quality Index
SWE = Scale of General Expectations of self-efficacy
CG = Control Group, EG = Experimental Group
d.f. = degree of freedom
SF-36 = Short Form 36 Health Survey Questionnaire
RP = Role of limitation physical functioning
GH = General Health
SF = Social functioning
EWB = Emotional well-being
IDS-SR = Self-Assessment Inventory of Depressive Symptoms
MCTQ = Munich Chronotype Questionnaire
MTVS = Mealtime Variability Score

Table S3 (continued)

Fig.	Variable	Group	Variable values (mean ± SD)				n-value	Statistical test	d.f.	F/t-value	Significance		Bonferroni post-hoc			
			Baseline T0	Explor. end T1	Interv. end T2	Follow-up T3					p-value	Summary	T0 vs. T1	T1 vs. T2	T2 vs. T3	
1E	BW	CG	.0 ± 0	-.1697 ± .9777	-.5606 ± 1.720		33	RM 1-way-ANOVA	2, 32	2.69	.0918	n.s.	.6524	n.s.	.2734	n.s.
		EG	.0 ± 0	-.2104 ± 1.048	-.2621 ± 2.226		67		2, 66	83.76	<.0001	****	.2099	n.s.	<.0001	****
	BW_c BW_d	EG			-3.379 ± 2.100	-4.500 ± 2.696	34	paired t-test	33	6.369	<.0001	****				
					-2.077 ± 2.110	-1.597 ± 2.152	30		29	3.487	.0016	**				
S2B	BMI	CG	.0 ± 0	-.06364 ± .3219	-.1970 ± .5537		33	RM 1-way-ANOVA	2, 32	3.12	.0658	n.s.	.5292	n.s.	.2418	n.s.
		EG	.0 ± 0	-.0620 ± .3446	-.8663 ± .7376		67		2, 66	83.75	<.0001	****	.2913	n.s.	<.0001	****
	BMI_c BMI_d	EG			-1.244 ± .6465	-1.728 ± .7676	34	paired t-test	33	6.072	<.0001	****				
					-.7033 ± .7289	-.5392 ± .7147	30		29	3.433	.0018	**				

SD = Standard deviation
CG = Control Group, EG = Experimental Group
BW = body weight
BMI = Body mass index
RM = Repeated Measure
_c = continuing
_d = discontinuing
d.f. = degree of freedom
n.s. = not significant

Table S4

Statistical data from the development of normalized body weight and BMI of participants during the exploration and the intervention phase, and of follow-up data of continuing and discontinuing participants. At T3, three participants could no longer be reached, which is why the n-number deviates here. The data in the table refers to the data shown in **Figures 1E** and **S2A**.

Fig.	Body weight change (mean ± SD)																Statistical test	ANOVA results	Source of Variation			Bonferroni post-hoc							
	T0				T1				T2				T3						Time	Group	Time x Group	T1		T2		T3			
	CG	n-value	EG	n-value	CG	n-value	EG	n-value	CG	n-value	EG	n-value	CG	n-value	EG	n-value						p-val.	Sum.	p-val.	Sum.	p-val.	Sum.		
1F	.00 ±.00	33	.00 ±.00	67	-.170 ±.978	33	-.210 ± 1.048	67	-.561 ± 1.720	33	-2.621 ± 2.226	67	-.796 ± 1.992	26	-4.500 ± 2.696	34	Mixed-effects model	d.f. F-value p-value Summary	1.6, 131.6 61.18 <.0001 ****	1, 98 35.51 <.0001 ****	3, 254 29.56 <.0001 ****	>.9999	n.s.	<.0001	****	<.0001	****		
S2C	.00 ±.00	33	.00 ±.00	67	-.064 ±.322	33	-.060 ±.346	67	-.197 ±.554	33	-.875 ±.735	67	-.273 ±.640	26	-1.479 ±.865	34	Mixed-effects model	d.f. F-value p-value Summary	1.7, 140.4 62.88 <.0001 ****	1, 98 35.95 <.0001 ****	3, 254 29.72 <.0001 ****	>.9999	n.s.	<.0001	****	<.0001	****		
S2G	.00 ±.00	33	.00 ±.00	67	-.170 ±.978	33	-.210 ± 1.048	67	-.367 ± 1.58	42	-2.270 ± 2.22	79					Mixed-effects model	d.f. F-value p-value Summary	1.4, 137.9 35.35 <.0001 ****	1, 119 11.14 .0011 **	2, 196 19.62 <.0001 ****	>.9999					n.s.	<.0001	****
	.00 ±.00	33	.00 ±.00	67	-.064 ±.322	33	-.060 ±.346	67	-.13 ±.513	42	-.759 ±.735	79					Mixed-effects model	d.f. F-value p-value Summary	1.4, 138 36.63 <.0001 ****	1, 119 10.88 .0013 **	2, 196 20.17 <.0001 ****								

SD = Standard deviation
CG = Control Group, EG = Experimental Group
d.f. = degree of freedom
n.s. = not significant

Table S5

Statistical data from the development of normalized body weight and BMI of participants at the different phases of the study. At T3 only continuing participants were included, which is why the n-number deviates here from the other time points. The data in the table refers to the data shown in **Figures 1F**, S2C, and S2G.

Fig.	Group	Variables	n-value	Statistical test	Slope ± SD	R ²	Is the slope significantly non-zero?						
							F-value	d.f.	p-value	Summary			
1G	CG	Δ MTVS vs. Δ body weight	33	Linear Regression	-.03350 ± .3929	.0002345	.007273	1, 31	.9326	n.s.			
		Δ MTVS vs. Δ BMI	33		-.06239 ± .1099	.01029	.5743	1, 31	.5743	n.s.			
	EG	Δ MTVS vs. Δ body weight	67		.8064 ± .2802	.1130	8.284	1, 65	.0054	**			
		Δ MTVS vs. Δ BMI	67		.2763 ± .09323	.1190	8.782	1, 65	.0042	**			
S2H	EG	Δ MTVS Breakfast vs. body weight	67	Linear Regression	-.1995 ± .1439	.02874	1.923	1, 65	.1702	n.s.			
		Δ MTVS Breakfast vs. BMI	67		-.07272 ± .04790	.03425	2.305	1, 65	.1338	n.s.			
		Δ MTVS Lunch vs. body weight	67		.6103 ± .2122	.1192	8.273	1, 65	.0054	**			
		Δ MTVS Lunch vs. BMI	67		.2050 ± .07080	.1142	8.380	1, 65	.0052	**			
		Δ MTVS Dinner vs. body weight	67		.5944 ± .2036	.1159	8.520	1, 65	.0048	**			
		Δ MTVS Dinner vs. BMI	67		.2067 ± .06762	.1257	9.342	1, 65	.0032	**			
S2I	EG	Baseline BMI vs. Δ BMI	67	Linear Regression	-.03515 ± .01779	.05974	4.130	1, 65	.0465	*			
2C	CG	Δ KJ intake vs. Δ body weight	30	Linear Regression	6.526e ⁻⁵ ± .0003174	.0001508	.04228	1, 28	.8386	n.s.			
		Δ KJ intake vs. Δ BMI	30		1.302e ⁻⁵ ± .0001046	.0005529	.01549	1, 28	.9018	n.s.			
	EG	Δ KJ intake vs. Δ body weight	67		-4.326e ⁻⁵ ± .00022	.0005975	.03886	1, 65	.8443	n.s.			
		Δ KJ intake vs. Δ BMI	67		-3.172e ⁻⁵ ± 7.319e ⁻⁵	.002881	.1878	1, 65	.6662	n.s.			
S2L	CG	KJ Exploration vs. KJ Intervention	30	Linear Regression	.8787 ± .1271	.6305	47.78	1, 28	<.0001	****			
	EG		67		.7094 ± .06780±	.6275	109.5	1, 65	<.0001	****			
2D	CG	Δ cumulative KJ intake vs. Δ body weight	30	Linear Regression	1.205e ⁻⁶ ± 7.270e ⁻⁶	.0009794	.02745	1, 28	.8696	n.s.			
		Δ cumulative KJ intake vs. Δ BMI	30		2.604e ⁻⁷ ± 2.395e ⁻⁶	.0004219	.01182	1, 28	.9142	n.s.			
	EG	Δ cumulative KJ intake vs. Δ body weight	67		-1.627e ⁻⁵ ± 5.237e ⁻⁶	.001483	.09655	1, 65	.7570	n.s.			
		Δ cumulative KJ intake vs. Δ BMI	67		-9.296e ⁻⁷ ± 1.746e ⁻⁵	.004341	.2834	1, 65	.5963	n.s.			
2F	CG	Fat	Δ intake vs. Δ body weight	30	Linear Regression	-.02193 ± .2513	.0002719	.007615	1, 28	.9311	n.s.		
			Δ intake vs. Δ BMI	30		-.001989± .08280	2.062e ⁻⁵	.0005774	1, 28	.9810	n.s.		
		EG	Δ intake vs. Δ body weight	67		.07281 ± .1600	.003176	.2071	1, 65	.6506	n.s.		
			Δ intake vs. Δ BMI	67		.02129 ± .05344	.002436	.1587	1, 65	.6916	n.s.		
	CG	Carb	Δ intake vs. Δ body weight	30		-.03933 ± .1090	.004631	.1303	1, 28	.7209	n.s.		
			Δ intake vs. Δ BMI	30		-.01866 ± .03580	.009609	.2717	1, 28	.6063	n.s.		
		EG	Δ intake vs. Δ body weight	67		-.1421 ± .08956	.03728	2.517	1, 65	.1175	n.s.		
			Δ intake vs. Δ BMI	67		-.04338 ± .03000	.03116	2.091	1, 65	.1530	n.s.		
	CG	Prot	Δ intake vs. Δ body weight	30		.02385 ± .2127	.0004489	.01257	1, 28	.9115	n.s.		
			Δ intake vs. Δ BMI	30		.02433 ± .06992	.004307	.1211	1, 28	.7304	n.s.		
		EG	Δ intake vs. Δ body weight	67		-.3055 ± .1843	.04054	2.747	1, 65	.1023	n.s.		
			Δ intake vs. Δ BMI	67		-.1050 ± .06147	.04293	2.916	1, 65	.0925	n.s.		
S2N	CG	Fat	Δ intake vs. Δ body weight	30	Linear Regression	-.000138 ± .000476	.002939	.08254	1, 28	.7760	n.s.		
			Δ intake vs. Δ BMI	30		-3.246e ⁻⁵ ± .000157	.001526	.04279	1, 28	.8376	n.s.		
	EG	Fat	Δ intake vs. Δ body weight	67		-8.216e ⁻⁵ ± .000319	.001021	.06640	1, 65	.7975	n.s.		
			Δ intake vs. Δ BMI	67		-4.923e ⁻⁵ ± .000106	.003286	.2143	1, 65	.6450	n.s.		
	CG	Carb	Δ intake vs. Δ body weight	30		9.359e ⁻⁵ ± .000181	.008762	.2475	1, 28	.6227	n.s.		
			Δ intake vs. Δ BMI	30		1.814e ⁻⁵ ± 6.214e ⁻⁵	.003034	.08522	1, 28	.7725	n.s.		
	EG	Carb	Δ intake vs. Δ body weight	67		-.000212 ± .000141	.03345	2.249	1, 65	.1385	n.s.		
			Δ intake vs. Δ BMI	67		-7.545e ⁻⁵ ± 4.712e ⁻⁵	.03796	2.565	1, 65	.1141	n.s.		
	CG	Prot	Δ intake vs. Δ body weight	30		-.000119 ± .000666	.001131	.03170	1, 28	.8600	n.s.		
			Δ intake vs. Δ BMI	30		3.299e ⁻⁵ ± .000219	.0008070	.02261	1, 28	.8815	n.s.		
			EG	Δ intake vs. Δ body weight		67	-.000806 ± .000449	.04714	3.216	1, 65	.0776	n.s.	
				Δ intake vs. Δ BMI		67	-.000281 ± .000150	.05154	3.532	1, 65	.0647	n.s.	
3B	CG	Δ 95%ile interval vs. Δ body weight	32	Linear Regression	.2213 ± .1476	.06972	2.248	1, 30	.1442	n.s.			
		Δ 95%ile interval vs. Δ BMI	32		.04259 ± .04231	.03268	1.013	1, 30	.3221	n.s.			
	EG	Δ 95%ile interval vs. Δ body weight	67		-.04842 ± .1218	.002426	.1581	1, 65	.6922	n.s.			
		Δ 95%ile interval vs. Δ BMI	67		-.02160 ± .04063	.004328	.2825	1, 65	.5968	n.s.			
3C	CG	Δ Breakfast time vs. Δ body weight	33	Linear Regression	.05080 ± .2608	.001223	.03794	1, 31	.8468	n.s.			
		Δ Breakfast time vs. Δ BMI	33		-.007265 ± .07335	.0003164	.009811	1, 31	.9217	n.s.			
	EG	Δ Breakfast time vs. Δ body weight	67		.1142 ± .1833	.005937	.3882	1, 65	.5354	n.s.			
		Δ Breakfast time vs. Δ BMI	67		.03573 ± .06123	.005210	.3404	1, 65	.5616	n.s.			
3D	CG	Δ Dinner time vs. Δ body weight	32	Linear Regression	.3149 ± .3682	.02380	.7313	1, 31	.3992	n.s.			
		Δ Dinner time vs. Δ BMI	32		-.01494 ± .1047	.0006776	.02034	1, 31	.8875	n.s.			
	EG	Δ Dinner time vs. Δ body weight	67		-.1096 ± .2361	.003308	.2157	1, 65	.6439	n.s.			
		Δ Dinner time vs. Δ BMI	67		-.05984 ± .07860	.008838	.5796	1, 65	.4492	n.s.			
3E	EG	midSleep Phase vs. midEat Phase	65	Linear Regression	.5902 ± .1449	.2084	16.58	1, 63	.0001	****			
3G	EG	Δ Residuals vs. Δ BMI	65	Linear Regression	.6532 ± .1176	.3286	30.83	1, 63	<.0001	****			
3J	EG	Correction of weekend meals vs. Δ BMI	mSP-mEP relation	55	Linear Regression	-.2522 ± .09116	.1084	7.656	1, 63	.0074	**		
			Breakfast	55		.1427 ± .08443	.05114	2.856	1, 53	.0969	n.s.		
			Lunch	64		.0460 ± .06415	.008225	.5142	1, 62	.4760	n.s.		
S2O	EG	Dev. weekend and scheduled meals	Dinner	63	Linear Regression	-.2490 ± .08055	.1355	9.558	1, 61	.0030	**		
			Breakfast	55		-.1174 ± .1031	.02391	1.298	1, 53	.2597	n.s.		
			Lunch	64		-.1409 ± .08067	.04690	3.051	1, 62	.0857	n.s.		
S2L	CG	KJ Exploration vs. KJ Intervention	Dinner	63	Linear Regression	-.03707 ± .1313	.001305	.07969	1, 61	.7787	n.s.		

BMI = Body mass index

CG = Control Group, EG = Experimental Group

MTVS = Mealtime Variability Score

d.f. = degree of freedom

n.s. = not significant

Table S6

Statistical data from all simple regression. The data in the table refers to the data shown in **Figs. 3A, B, and C**. In data for **Figures 2B, D, and S2H**, Some participants were excluded from this analysis because of several probably erroneous diary entries about unrealistic amounts of food. In data for **Figures 3B-D**, varying n-numbers result from incomplete entries in the diaries. In data for **Figures 3E, G, J, and S2I**, missing n-numbers are due to incomplete diary entries on weekends during the exploratory phase or because participants usually did not have a particular meal.

Weekly MTVS	Corresponding time window (min)	Weekly weight change (kg)	SD	n-value	Sig. difference from MTVS =1
= 1	≤ 30	-0.47	0.1995	50	
≤ 2	≤ 60	-0.39	0.5144	318	n.s.
≤ 3	≤ 90	-0.38	0.5881	426	n.s.
> 3	> 90	-0.12	0.5415	280	*
> 4	> 120	-0.10	0.4035	137	*

Table S7

Association of weekly MTVS with weekly weight loss. The table shows the weekly weight change of all participants during the exploration and the intervention phase while adhering to certain MTVSs in each week. 1-way ANOVA with Bonferroni post-hoc test comparing the mean of each row with the mean of the first row, * $p \leq 0.05$.

Fig.	BMI change (mean ± SD)												Statistical test	ANOVA results	Source of Variation			Bonferroni post-hoc					
	20.0-24.9				25.0-29.9				>30									20.0-24.9		25.0-29.9		>30	
	CG	n-value	EG	n-value	CG	n-value	EG	n-value	CG	n-value	EG	n-value					Weight	Group	Weight x Group	p-val.	Sum.	p-val.	Sum.
S2J	-.018 ± .396	17	-.742 ± .546	26	-.255 ± .468	11	-.776 ± .709	24	-.304 ± .728	5	-.951 ± .804	17	2-way ANOVA	d.f.	2, 94	1, 94	2, 94	<.0001	***	.0694	n.s.	.1295	n.s.
													F-value	1.012	19.03	.2299							
													p-value	.3673	<.0001	.7950							
													Summary	n.s.	****	n.s.							

SD = Standard deviation

CG = Control Group, EG = Experimental Group

d.f. = degree of freedom

n.s. = not significant

Table S8

Statistical data from BMI changes depending of the baseline BMI of each participant. The data in the table refers to the data shown in [Figure S2J](#).

Fig.	Model summary		Model		Coefficients						
			Analysis of Variance	F (DfN, DfD)	Parameter estimates	Variable	Estimate	Stand. Err.	t	P value	Summary
1H	R ²	.1518	Regression	F (2, 64) = 5.729	β0	Intercept	.7183	.5139	1,398	.1670	n.s.
			D_MTVS	F (1, 64) = 6.949	β1	D_MTVS	.2438	.09248	2,636	.0105	*
			Pre_BMI	F (1, 64) = 4.439	β2	Pre_BMI	-.03588	.01703	2,107	.0391	*
3K	R ²	.3970	Regression	F (6, 45) = 4.939	β0	Intercept	-.1691	.2434	.6947	.4908	n.s.
			D_MTVS	F (1, 45) = 5.977	β1	D_MTVS	.2582	.1056	2,445	.0185	*
			D_KJ	F (1, 45) = .1534	β2	D_KJ	2,783 ^{e-5}	7,105e-005	.3917	.6971	n.s.
			Delta Intake Interval	F (1, 45) = 0.6980	β3	Delta Intake Interval	-.04606	.05513	.8355	.4079	n.s.
			Delta Residuals	F (1, 45) = 3.232	β4	Delta Residuals	-.2217	.1233	1,798	.0789	n.s.
			Breakfast Jetlag	F (1, 45) = 3.012	β5	Breakfast Jetlag	.1457	.08395	1,736	.0895	n.s.
			Dinner Jetlag	F (1, 45) = 13.01	β6	Dinner Jetlag	-.3071	.08513	3,608	.0008	***

BMI = Body mass index

MTVS = Mealtime variability score

d.f. = degree of freedom

Reg. Coeff = Regression coefficient

Std. Err. = Standard error

Table S9

Statistical data from multiple regression analysis testing the effects of different variables while keeping the others constant. The data in the table refers to the data shown in [Figures 1H](#) and [3K](#).

Fig.	Data set	Normalized midSleep Phase	Normalized midEat Phase					n-value
			Relative Phase	SD	Vector length	Rayleigh Z-value	p-value	
3F	Exploration weekend	.00	11.33	20.653°	.937	57.093	<1e ⁻¹²	65
	Intervention	.00	10.57	17.221°	.956	60.299	<1e ⁻¹²	65

SD = Standard deviation

Table S10

Statistical data from Rayleigh's uniformity test analyzing phase distribution from the relation between midSleep Phase and midEat Phase. Missing n-numbers are due to incomplete diary entries on weekends during the exploratory phase. The data in the table refers to the data shown in [Fig. 3F](#).

Fig.	Meal	Variable value (mean ± SD)		n-value	Statistical test	d.f.	t-value	Significance	
		Exploration phase T0-T1	Intervention phase T1-T2					p-value	Summary
3I	Breakfast	.0 ± .0	-.6477 ± 1.120	55	paired t-test	54	4.288	<.0001	****
	Lunch	.0 ± .0	-.6384 ± 1.346	64		63	3.796	.0003	***
	Dinner	.0 ± .0	-.0002648 ± 1.013	63		62	.0002648	.9984	n.s.

SD = Standard deviation

d.f. = degree of freedom

n.s. = not significant

Table S11

Statistical data on shifts in weekend eating times caused by the change to calculated eating times in the intervention phase. Missing n-numbers are due to incomplete diary entries on weekends during the exploratory phase or because participants usually did not have a particular meal. The data in the table refers to the data shown in [Fig. 3I](#).

Construct	Questionnaire	Components
Sleep Quality	Pittsburgh Sleep Quality Index (PSQI)	Subjective sleep quality, sleep latency, sleep duration, habitual sleep efficiency, sleep disturbances, daytime impairment.
Physical Well-Being	36-Item Short Form Health Survey (SF-36)	Vitality, physical functionality, physical pain, general health perception, physical role function, emotional role function, social function, mental well-being.
Self-Efficacy	Scale of General Expectations of Self-Efficacy (SWE)	Optimistic competence expectation, daring to solve life problems independently, attribution of success to own competences.
Depressive Mood	Self-Assessment Inventory of Depressive Symptoms (IDS-SR)	Sleep quality, mood, appetite, weight changes, concentration/decision making, self-esteem, future expectations, suicidal thoughts, activities, energy levels, pleasure/joy, sexual interest, slowing down, restlessness, physical/vegetative complaints, panic/anxiety, digestive problems, interpersonal sensitivity, physical heaviness.
Chronotype	Munich Chronotype Questionnaire (MCTQ)	Bedtimes, sleep times, energy levels, exposure to natural light, differences between weekdays and weekends.

Table S12

Questionnaires collected at T0 and T2 to assess general well-being of participants.

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Reviewer #1 (Public Review):

The authors Wilming and colleagues set out to determine the impact of regularity of feeding per se on the efficiency of weight loss. The idea was to determine if individuals who consume 2-3 meals within individualized time frames, as opposed to those who exhibit stochastic feeding patterns throughout the circadian period, will cause weight loss.

The methods are rigorous, and the research is conducted using a two-group, single-center, randomized-controlled, single-blinded study design. The participants were aged between 18 and 65 years old, and a smartphone application was used to determine preferred feeding times, which were then used as defined feeding times for the experimental group. This adds strength to the study since restricting feeding within preferred/personalized feeding windows will improve compliance and study completion. Following a 14-day exploration phase and a 6-week intervention period in a cohort of 100 participants (inclusive of both the controls and the experimental group that completed the study), the authors conclude that when meals are

restricted to 45min or less durations (MTVS of 3 or less), this leads to efficient weight loss. Surprisingly, the study excludes the impact of self-reported meal composition on the efficiency of weight loss in the experimental group. In light of this, it is important to follow up on this observation and develop rigorous study designs that will comprehensively assess the impact of changes (sustained) in dietary composition on weight loss. The study also reports interesting effects of regularity of feeding on eating behavior, which appears to be independent of weight loss. Perhaps the most important observation is that personalized interventions that cater to individual circadian needs will likely result in more significant weight loss than when interventions are mismatched with personal circadian structures. One area of concern for the study is its two-group design; however, single-group cross-over designs are tedious to develop, and an adequate 'wash-out' period may be difficult to predict. A second weakness is not considering the different biological variables and racial and ethnic diversity and how that might impact outcomes. In sum, the authors have achieved the aims of the study, which will likely help move the field forward.

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Reviewer #2 (Public Review):

Summary:

The authors investigated the effects of the timing of dietary occasions on weight loss and well-being with the aim of explaining if a consistent, timely alignment of dietary occasions throughout the days of the week could improve weight management and overall well-being. The authors attributed these outcomes to a timely alignment of dietary occasions with the body's own circadian rhythms. However, the only evidence the authors provided for this hypothesis is the assumption that the individual timing of dietary occasions of the study participants identified before the intervention reflects the body's own circadian rhythms. This concept is rooted in understanding of dietary cues as a zeitgeber for the circadian system, potentially leading to more efficient energy use and weight management. Furthermore, the primary outcome, body weight loss, was self-reported by the study participants.

Strengths:

The innovative focus of the study on the timing of dietary occasions rather than daily energy intake or diet composition presents a fresh perspective in dietary intervention research. The feasibility of the diet plan, developed based on individual profiles of the timing of dietary occasions identified before the intervention, marks a significant step towards personalised nutrition.

Weaknesses:

Several methodological issues detract from the study's credibility, including unclear definitions not widely recognized in nutrition or dietetics (e.g., "caloric event"), lack of comprehensive data on body composition, and potential confounders not accounted for (e.g., age range, menstrual cycle, shift work, unmatched cohorts, inclusion of individuals with normal weight, overweight, and obesity). The primary outcome's reliance on self-reported body weight and subsequent measurement biases further undermines the reliability of the findings. Additionally, the absence of registration in clinical trial registries, such as the EU Clinical Trials Register or clinicaltrials.gov, and the multiple testing of hypotheses which were not listed a priori in the research protocol published on the German Register of Clinical Trials impede the study's transparency and reproducibility.

Achievement of Objectives and Support for Conclusions:

The study's objectives were partially met; however, the interpretation of the effects of meal timing on weight loss is compromised by the weaknesses mentioned above. The evidence only partially supports some of the claims due to methodological flaws and unstructured data analysis.

Impact and Utility:

Despite its innovative approach, significant methodological and analytical shortcomings limit the study's utility. If these issues were addressed, the research could have meaningful implications for dietary interventions and metabolic research. The concept of timing of dietary occasions in sync with circadian rhythms holds promise but requires further rigorous investigation.

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Reviewer #3 (Public Review):

The authors tested a dietary intervention focused on improving meal regularity in this interesting paper. The study, a two-group, single-center, randomized, controlled, single-blind trial, utilized a smartphone application to track participants' meal frequencies and instructed the experimental group to confine their eating to these times for six weeks. The authors concluded that improving meal regularity reduced excess body weight despite food intake not being altered and contributed to overall improvements in well-being.

The concept is interesting, but the need for more rigor is of concern.

A notable limitation is the reliance on self-reported food intake, with the primary outcome being self-reported body weight/BMI, indicating an average weight loss of 2.62 kg. Despite no observed change in caloric intake, the authors assert weight loss among participants.

The trial's reliance on self-reported caloric intake is problematic, as participants tend to underreport intake; for example, in the NEJM paper (DOI: 10.1056/NEJM199212313272701), some participants underreported caloric intake by approximately 50%, rendering such data unreliable and hence misleading. More rigorous methods for assessing food intake are available and should have been utilized. Merely acknowledging the unreliability of self-reported caloric intake is insufficient as it would still leave the reader with the impression that there is no change in food intake when we actually have no idea if food intake was altered. A more robust approach to assessing food intake is imperative. Even if a decrease in caloric intake is observed through rigorous measurement, as I am convinced a more rigorous study would unveil testing this paradigm, this intervention may merely represent another short-term diet among countless others that show that one may lose weight by going on a diet, principally due to heightened dietary awareness.

Furthermore, the assessment of circadian rhythm using the MCTQ, a self-reported measure of chronotype, may not be as reliable as more objective methods like actigraphy.

Given the potential limitations associated with self-reported data in both dietary intake and circadian rhythm assessment, the overall impact of this manuscript is low. Increasing rigor by incorporating more objective and reliable measurement techniques in future studies could strengthen the validity and impact of the findings.

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